

Arithmetic Continuations and Local Certificates for Unknot Recognition

An implementation, obstruction theorems,
and a route toward quasi-polynomial complexity

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Scope. This report contributes checked arithmetic recognition for supplied Montesinos presentations, verified local obstructions in ordinary planar diagrams, explicit continuation and chain-complex lower bounds, reproducible software, and a restricted quasi-polynomial witness-discovery theorem. It does not establish a general quasi-polynomial running-time bound for the ProveIt recognizer.

Repository baseline. <https://github.com/VladimirReshetnikov/ProveItTopology/UnknotRecognition>
Commit 86f23490006c880cd01db9bec10761d39c6395c8.

Research status. Classical results are attributed explicitly. New statements below are derivations for this project, with complete mathematical arguments and independent computational checks. They are not asserted to be first-in-literature results or machine-checked formal proofs.

Source, software patch, certificates, tests, and raw measurements
are distributed together with this report.

Abstract

We develop an arithmetic alternative to expanding every intermediate tangle complex in exact unknot recognition. A four-ended family of rational prefixes and rational continuations has a Boolean unknot-recognition connection matrix equal to an identity matrix of order 2^m , although the signed gluing-value matrix has rank at most two and is represented by integers with $O(m)$ bits. Thus exponentially many exact continuation profiles coexist with polynomial-time arithmetic evaluation. On a related family of actual unknot diagrams, preserving a complete explicit four-ended Bar–Natan complex at a specified prefix requires exponentially many summands. These statements identify precise limitations of finite profile tables and full-complex expansion; neither is a lower bound for general unknot decision time.

We implement a complete classifier for supplied finite-slope Montesinos presentations using exact continued-fraction arithmetic, component checks, and a nontrivial triangle-group quotient of the branched cover. The production arithmetic has quadratic schoolbook bit complexity in the compressed source length. A separate verifier and checked diagram compiler make the input contract explicit. A second implementation searches ordinary planar diagrams for small Montesinos tangles that cannot embed in any unknot, using the Nogueira–Salgueiro criterion and a verified separating-disk certificate. Enumerating all such literal templates with $O(\log n)$ coefficients gives a proved $n^{O(\log n)}$ algorithm for that restricted discovery problem. This exhaustive enumerator is specified and proved, while the delivered implementation searches a fixed or supplied finite catalogue.

Paired measurements on the supplied-presentation family show large recognition-stage improvements and resolve examples that exceed an explicit complex-object cap in the old path. Local catalogue search also has measured regressions and remains optional. We give a claim ledger, reproducible tests and timings, and a prioritized research programme addressing discovery, mixed arithmetic interfaces, homological information, and geometric hierarchy operation costs.

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1 Results, input models, and the remaining objective

The most useful next step for this project is to change which intermediate objects are materialized. The existing recognizer already avoids much unnecessary work through local simplification, algebraic obstructions, connected-sum structure, and improved Khovanov computations. Those changes do not make an explicit complex small merely because its boundary is small. The present contribution supplies an exact arithmetic route around that problem on a well-defined family, and a local certificate route that applies to a larger family of diagrams.

The central engineering recommendation is to retain certified rational pieces as primitive integer slopes and decide supported gluings arithmetically. When a general diagram contains a tangle that is incompatible with every unknotting complement, a checked local occurrence can end the decision without evaluating the exterior. Each recommendation has a precise input contract, a proof obligation, executable code, and a measured limitation.

1.1 What is established here

Table 1 distinguishes mathematical conclusions from implementation and discovery claims. This distinction is necessary because a fast algorithm on a supplied presentation is not automatically an algorithm that discovers that presentation in an arbitrary diagram.

Contribution	Exact scope	Status
Exponential continuation profiles	Four marked endpoints, $O(m)$ crossings per prefix and continuation, all cross closures knots	Proved; arithmetic diagnostic through $m = 10$
Exponential explicit prefix complex	Any full delooped Bar–Natan complex for a specified rational prefix inside an unknot family	Proved; not a scan-order-independent decision lower bound
Montesinos arithmetic classifier	Supplied finite-slope continued fractions and integer term; one-component input checked	Implemented, complete on this input class
Forbidden subtangle search	Literal connected disk patterns in a validated plane knot diagram; four distinct cut edges	Implemented for supplied or fixed finite catalogue
Quasi-polynomial discovery	Literal admissible patterns with at most $O(\log n)$ continued-fraction coefficients	Enumeration algorithm proved; exhaustive enumerator not implemented
General quasi-polynomial recognizer	Every valid n -crossing knot diagram	Not established by this contribution

Table 1: Claim ledger. *Proved* refers to the mathematical arguments in this report, rather than formal verification in a proof assistant.

1.2 Three input sizes that must be kept separate

Let n be the number of crossings of a supplied planar diagram (PD). Its four-valent plane map, crossing information, and component structure are part of the input. With canonically bounded labels its encoding has $O(n \log(n + 2))$ bits. If users supply labels with arbitrarily

many bits, their actual encoding length must also be charged to parsing. Word-operation bounds below assume $O(\log(n+2))$ -bit indices after relabeling.

Let B be the binary length of a structured tangle source. It includes the signed integer term, all signed continued-fraction coefficients, and list delimiters. Counting delimiters matters: a list containing many zeros can have almost no twists but still take time to read and evaluate. Let

$$C = |e| + \sum_{i,j} |a_{i,j}|$$

be the number of crossings in the literal expansion of that source. The compiler in this contribution constructs precisely that literal diagram, before any optional Reidemeister simplification. In general C can be exponential in B , so an $O(B^2)$ arithmetic result does not justify expanding a PD in $O(B^2)$ time.

Finally, let h denote the *number* of continued-fraction coefficients in a local template description. The discovery theorem in Section 4 assumes both a bound on h and an explicit crossing budget. It never equates coefficient count with bit length.

We use *quasi-polynomial* for a bound $\exp((\log n)^{O(1)})$. The more specific target $n^{O(\log n)} = \exp(O((\log n)^2))$ is stronger than $n^{O((\log n)^2)} = \exp(O((\log n)^3))$. Whenever an exponent matters we write it explicitly.

1.3 The audited repository baseline

The work starts from ProveIt commit `86f23490006c880cd01db9bec10761d39c6395c8`, retrieved on 8 October 2026 [1]. The audited executable path is the `fast` subdirectory under `Topology/UnknotRecognition`. Its exact source and relevant tests, examples, and recent research documents were retrieved through the GitHub interface. Git blob hashes were checked against the fetched bytes for 82 files; the source manifest accompanies the integration package. The baseline test suite contains 134 tests, despite the older test count still present in an earlier README passage.

The baseline already includes Reidemeister reduction, descending-diagram recognition, complete specialized three-braid decisions, signed Seifert and Rasmussen-style obstructions, modular and exact Alexander paths, a Jones filter, visible connected-sum and interlacement decomposition, and several exact Khovanov backends. Component sharing, saturated multiplicities, Euler bounds, sparse maps, and composition caching are existing work. The recent reports additionally discuss radical minimization, exact homological windows, continuation shortening, Fitting splits, terminal boundary tests, twist macros, and hierarchy restart counts. These are context for the present work and are not relabeled as new contributions.

The new production stage is deliberately earlier than complex construction when a checked rational source exists. Its mathematical information is different from an approximate Jones or Alexander value: within the supported Montesinos class it is a complete decision. The general exact fallback continues to have an exponential worst-case upper bound and a finite-budget UNKNOWN result. The JSON flag reporting the absence of a general quasi-polynomial guarantee is retained.

1.4 Relevant literature and the status of the broad target

The exact Khovanov decision criterion ultimately uses the theorem of Kronheimer and Mrowka that reduced Khovanov homology detects the unknot [2]. Over $\mathbb{F}_2$, a reduced rank of one also forces rational reduced rank one by the universal coefficient inequality. This supplies a sound decision criterion; it does not give a fast method for computing the required rank.

Lackenby’s March 2021 Oxford slides announce a general unknot algorithm with running time $2^{O((\log n)^3)}$ [3]. The power three in the logarithm should not be silently replaced by two. The July 2026 manuscript *Incompressible surfaces, hierarchies and unknot recognition* gives a new hierarchy algorithm and substantial topological structure [4]. Section 9 of that manuscript explicitly separates an iteration bound from a full running-time analysis because several operations have not been specified sufficiently for the latter. Its displayed iteration estimate is

$$L(g+1)^L,$$

where L is the maximum hierarchy length and g bounds surface pattern-complexity. The refined construction in Section 10 controls hierarchy length in terms of the initial handle complexity.

This report uses that distinction constructively. A project implementation needs explicit operation algorithms, encoding-size bounds, and the appropriate logarithmic hierarchy estimate before inheriting a quasi-polynomial bound. A 2021 announcement, a 2026 iteration estimate, and a proved executable bound for the present code are different kinds of evidence. We do not infer the last from either of the first two. This is a statement about the sources and implementation audited here, not a claim to have ruled out every other possible algorithm in the literature.

The rational-tangle classification and gluing formulas used below are classical; we use the treatments of Kauffman and Lambropoulou [5, 6, 7]. The local obstruction is the theorem of Nogueira and Salgueiro [9]. The lower bounds in the next sections are project-specific deductions from these tools and the functorial Khovanov framework, with their own proofs.

1.5 Use of the suggested mathematics repository

The suggested `openai/math` repository was inspected at the following commit:

`adc7f1241b42e322a6451854ab7e4b4c146bf78a.`

Its description distinguishes research manuscripts at different stages from formally verified results. The relevant inspiration was a standard context-separation proof pattern: equal summaries must give equal answers after every allowed continuation. The inspected section applies this idea to word relations and automata [14].

Our rational-tangle construction below is independent of that manuscript’s headline automata claim. It specifies actual knot diagrams, proves the diagonal acceptance matrix using unimodular arithmetic and disjoint cones, and gives a separate arithmetic representation. No theorem from the suggested repository is used as a black box for topology or complexity. The exact source path, commit, and blob hash are recorded in the package’s research provenance.

2 Four-ended rational tangles have exponentially many decision profiles

The following construction separates three different size measures: the number of exact continuation-equivalence classes, the dimension of a linear representation of the final Boolean answer, and the number of bits needed for an arithmetic representation. The first two are exponential even at a fixed four-point boundary. The third is linear on the same family, and the resulting decision procedure has polynomial bit complexity.

2.1 Rational tangle conventions and the gluing test

Write $T(p/q)$ for a rational two-string tangle with reduced fraction p/q , and write $N(T)$ for numerator closure. All four boundary points are marked. A minus sign on a tangle denotes its mirror image. We use

$$F(T + [k]) = F(T) + k, \quad F(T^{\text{rot}}) = -\frac{1}{F(T)}.$$

Horizontal addition of two crossings and vertical addition of two crossings therefore act on a primitive fraction column by

$$A \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} p + 2q \\ q \end{pmatrix}, \quad B \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} p \\ 2p + q \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}. \quad (2.1)$$

Here the vertical operation is $T \mapsto T * 1/[2] = 1/([2] + 1/T)$. Each operation adds two crossings and preserves the pairing of the four marked endpoints. These fraction and tangle operations are reviewed in [7, Section 2].

We use the following classical, exact rational-tangle gluing criterion:

$$N(T(p/q) - T(r/s)) \text{ is the unknot} \iff |ps - qr| = 1. \quad (2.2)$$

This is [7, Theorem 5], with the second fraction negated. The two input fraction pairs must be primitive separately. The integer $ps - qr$ is computed before any cancellation in the sum of the fractions. For example, $1/2 + 1/2 = 1$ as a rational number, whereas the numerator closure of $T(1/2) + T(1/2)$ is not an unknot: the integer in the sum version of the criterion is $1 \cdot 2 + 2 \cdot 1 = 4$. This distinction is necessary because a sum of rational tangles need not itself be a rational tangle.

2.2 A family of disjoint positive cones

Let $\mathcal{C} = \mathbb{R}_{>0}^2$. For a word $w = w_1 \cdots w_m \in \{A, B\}^m$, let

$$G_w = w_1 \cdots w_m \in \text{SL}_2(\mathbb{Z}), \quad \mathcal{C}_w = G_w \mathcal{C}.$$

The product convention is literal matrix multiplication. To construct the corresponding tangle from its base tangle, apply the letters of w from right to left.

Lemma 2.1 (Disjoint cones). *If $u, w \in \{A, B\}^m$ are distinct, then $\mathcal{C}_u \cap \mathcal{C}_w = \emptyset$. Moreover, if $z \in \mathcal{C}_u$ and $y = G_w^{-1}z \in \mathbb{Z}^2$, then at least one coordinate of y is nonpositive and at least one coordinate is strictly positive.*

Proof. Every generator preserves $\mathcal{C}$, and

$$AC = \{(x, y) : y > 0, x > 2y\}, \quad BC = \{(x, y) : x > 0, y > 2x\}.$$

These two sets are disjoint. Remove the longest common prefix of u, w ; its matrix is invertible. The remaining two words start with different generators, and their tail matrices preserve $\mathcal{C}$. Their cone images are therefore contained in the two disjoint sets above. Applying the common prefix back proves disjointness.

If both coordinates of y were positive, then $z \in \mathcal{C}_w$, contradicting disjointness. If both were nonpositive, the nonnegative matrix G_w would send y to a vector with both coordinates nonpositive, contradicting $z \in \mathcal{C}_u \subset \mathcal{C}$. This proves the asserted sign pattern. It also handles a zero coordinate explicitly: the other coordinate must then be a positive integer. $\square$

Theorem 2.2 (An exponential identity submatrix using knots only). *For every integer $m \geq 0$, there are 2^m rational prefix tangles P_w and 2^m rational continuation tangles Q_w , each with the same four marked boundary points, such that:*

1. P_w has a diagram with at most $2m + 1$ crossings, and Q_w has a diagram with at most $2m + 3$ crossings;
2. every closure $K_{u,w} = N(P_u + Q_w)$ is a one-component knot;
3. $K_{u,w}$ is the unknot if and only if $u = w$.

In particular, the matrix

$$H_m(u, w) = \mathbf{1}\{N(P_u + Q_w) \text{ is the unknot}\}$$

is the $2^m \times 2^m$ identity matrix and has rank 2^m over every field. Every displayed closed diagram has at most $4m + 4$ crossings.

Proof. Define primitive columns

$$v_w = \begin{pmatrix} p_w \\ q_w \end{pmatrix} = G_w \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad x_w = \begin{pmatrix} r_w \\ s_w \end{pmatrix} = G_w \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

All coordinates are positive integers. Primitivity follows because the base columns are primitive and G_w is unimodular. Let

$$P_w = T(p_w/q_w), \quad Q_w = -T(r_w/s_w),$$

using the diagrams obtained from base tangles $T(1)$ and $T(3/2)$ by the two-crossing operations in (2.1). The bases have one and three crossings, respectively; the latter is $[1, 2] = [1] + 1/[2]$. Taking a mirror image preserves the crossing count. The asserted size bounds follow.

Each generator adds an even number of twists to an adjacent pair of boundary endpoints, so it preserves the endpoint pairing. Thus every P_w has the pairing of $T(1)$, and every Q_w has that of $-T(3/2)$. Rational tangles have two arcs and no closed components. Joining these two fixed endpoint pairings and adding the two numerator caps produces one circle; equivalently, the base closure $N(T(1) - T(3/2))$ is an unknot by (2.2) and all subsequent two-twist changes preserve its component count. Hence every cross closure is a knot. As an independent parity check, $G_w \equiv I \pmod{2}$, so

$$v_w \equiv (1, 1)^t, \quad x_w \equiv (1, 0)^t \pmod{2}, \quad \det(v_u, x_w) \equiv 1 \pmod{2}.$$

For the diagonal entries, unimodularity gives

$$\det(v_w, x_w) = \det\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix}\right) = -1.$$

For $u \neq w$, let $y = (a, b)^t = G_w^{-1}v_u$. By Lemma 2.1, a, b are integers with one nonpositive coordinate and one positive coordinate. Therefore

$$\det(v_u, x_w) = \det\left(\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix}\right) = 2a - 3b.$$

If $a \leq 0 < b$, this is at most -3 ; if $b \leq 0 < a$, it is at least 2. Consequently its absolute value is at least two. The parity calculation in fact makes it at least three. Criterion (2.2) proves the precise diagonal acceptance pattern. The rank assertion follows from the identity matrix. $\square$

Remark 2.3 (A direct numerical gap). The generators above give the stronger off-diagonal bound $|\det(v_u, x_w)| \geq 19$. Remove the longest common prefix. If the first different letters are A on the row side and B on the column side, write the remaining positive columns as $(a, b)^t$ and $(c, d)^t$. Here $a, b \geq 1$, $c \geq 3$, $d \geq 2$. Then

$$\det \left(A \begin{pmatrix} a \\ b \end{pmatrix}, B \begin{pmatrix} c \\ d \end{pmatrix} \right) = 2ac + ad + 3bc + 2bd \geq 21.$$

In the opposite order the determinant is $-(2ac + 3ad + bc + 2bd) \leq -19$. This gap is useful as a simple exact-arithmetic diagnostic; the cone argument alone already proves the theorem.

2.3 Consequences for continuation summaries

Corollary 2.4 (Finite-state and linear-summary obstructions). *Consider an exact summary $\sigma(P)$ of a four-ended prefix tangle from which the unknot decision can be recovered for every admitted continuation, and suppose the admitted continuations include the Q_w above. On prefixes with at most $2m + 1$ crossings, σ needs at least 2^m distinct values. Any fixed-length binary representation of these values needs at least m bits.*

If, more specifically, there are vectors $f(P) \in \mathbb{F}^d$ and linear functionals λ_Q satisfying

$$\lambda_Q(f(P)) = \mathbf{1}\{N(P + Q) \text{ is the unknot}\}$$

for every such prefix and continuation, then $d \geq 2^m$. The same conclusion holds for a bilinear factorization of the Boolean connection matrix through a d -dimensional vector space.

Proof. The continuation Q_w accepts P_w and rejects every other P_u in the family, so their summaries must all be distinct. The bit lower bound is $\log_2(2^m) = m$. In the linear case the restricted matrix H_m factors through a vector space of dimension d , so its rank is at most d . Apply Theorem 2.2. $\square$

This argument concerns exact representations of *all* continuation answers. It does not give a running-time lower bound for unknot recognition. In particular, 2^m different states can be represented by $O(m)$ bits. The four-ended interface is fixed throughout. Each tangle is built by twisting two of four boundary strands, and a collar decomposition gives constant-size intermediate boundaries. Any use of this observation for a particular PD sweep convention should state that convention explicitly.

Theorem 2.5 (A rank-two arithmetic representation with a nonlinear readout). *For rational tangles supplied by explicit two-twist words as above, the pair of integers (p, q) is an exact tangle summary using $O(m)$ bits. Two such summaries decide the closure unknot question in $O(m^2)$ bit operations using schoolbook arithmetic and $O(m)$ working bits.*

For the family in Theorem 2.2, the signed gluing-value matrix

$$D_m(u, w) = p_u s_w - q_u r_w$$

has rank at most two over $\mathbb{Q}$, whereas $H_m(u, w) = \mathbf{1}\{|D_m(u, w)| = 1\}$ has rank 2^m .

Proof. Both generators have infinity-operator norm at most three. Thus

$$\|v_w\|_\infty \leq 3^m, \quad \|x_w\|_\infty \leq 3^{m+1}.$$

Their coordinates have $O(m)$ binary digits. Each generator update is one doubling and one addition of integers having $O(m)$ bits. Performing all m updates uses $O(m^2)$ bit operations

and $O(m)$ working bits. The final two multiplications, subtraction and comparison with ± 1 also fit these bounds by schoolbook integer multiplication.

Conway's classification of rational tangles by their fraction [5, Theorem 1] makes (p, q) , normalized up to simultaneous sign, an exact summary of the rational tangle with marked boundary. Efficient evaluation of a gluing is supplied here for rational continuations by (2.2).

Finally, if p, q, r, s denote the column vectors of the corresponding coordinates across the finite family, then

$$D_m = ps^t - qr^t.$$

This is a sum of two rank-one matrices, so its rank is at most two. The Boolean-rank assertion is Theorem 2.2. $\square$

The new obstruction therefore identifies a limitation of one specific compression target: linear factorization of the final exact Boolean predicate. Keeping the signed arithmetic gluing value before applying its nonlinear readout avoids that obstruction on rational tangles. Extending this arithmetic representation beyond certified rational pieces requires additional topological structure and remains a research question.

3 An exponential full-complex checkpoint inside explicit unknots

The Boolean connection-rank obstruction concerns a representation that answers every admitted continuation. The next obstruction is different: it applies to the explicit size of a complex that retains the complete Bar–Natan homotopy type of one specified prefix. It holds even when the input is promised to be an unknot.

3.1 A reduced closure functor needs no exactness hypothesis

Fix a field $\mathbb{F}$ and the ordinary Khovanov Frobenius algebra $A = \mathbb{F}[X]/(X^2)$. Work in the additive Bar–Natan category with fixed boundary, using the standard specialization if a universal deformation is present. We count actual summands, with multiplicity, of circle-free crossingless matchings in a bounded, explicitly delooped complex. A quantum or homological shift does not change this count.

Lemma 3.1 (Closure lower bound). *Let P be a tangle with $2r > 0$ boundary points. Fix a crossingless closure Γ for which $\Gamma(P)$ is a knot. Suppose that C is homotopy equivalent to the full Bar–Natan complex of P and that all objects appearing in C are shifts of circle-free crossingless matchings. If $M(C)$ is the total number of summands over all chain degrees, counted with multiplicity, then*

$$M(C) \geq 2^{1-r} \dim_{\mathbb{F}} \widetilde{Kh}(\Gamma(P); \mathbb{F}) \geq 2^{1-r} \det(\Gamma(P)).$$

In particular, for a four-ended tangle and numerator closure, $M(C) \geq \det(N(P))/2$.

Proof. Choose a point $*$ in the interior of one of the external closure arcs. Closing a cobordism adds the fixed product $\Gamma \times [0, 1]$. Consequently the track $* \times [0, 1]$ lies on the resulting cobordism and joins its distinguished input and output circles. There is no requirement that the original tangle category carry a basepoint.

Let F be the ordinary Khovanov TQFT. Give $F(\Gamma(S))$ its A -module structure by multiplication on the circle containing $*$. Every map obtained by closing a tangle cobordism is A -linear: a multiplication by X can be moved along the fixed marked track from the incoming

end to the outgoing end. Algebraically this follows from commutativity of multiplication and the Frobenius identity

$$\Delta(Xa) = (X \otimes 1)\Delta(a) = (1 \otimes X)\Delta(a).$$

Births and deaths away from the marked track commute with this action as well. These identities apply to composite cobordisms and their linear combinations. The usual sphere, torus, and neck-cutting relations are already respected by F .

Thus the assignment

$$R_\Gamma(S) = F(\Gamma(S)) \otimes_A A/(X)$$

and the induced maps define an additive functor on the original, unpointed tangle category. Equivalently, up to a grading shift, one can restrict to the image of multiplication by X on the marked circle. All modules before reduction are free as A -modules, so the quotient and image descriptions are naturally identified by $[a] \mapsto Xa$.

An additive functor preserves the equations defining a chain homotopy. For example, $gf - 1 = dh + hd$ implies $R_\Gamma(g)R_\Gamma(f) - 1 = R_\Gamma(d)R_\Gamma(h) + R_\Gamma(h)R_\Gamma(d)$. It therefore preserves chain homotopy equivalences. No exactness of $-\otimes_A A/(X)$ is asserted or needed. Applied to the cube of resolutions of P , the functor gives the reduced Khovanov complex of $\Gamma(P)$, up to harmless grading shifts. This is compatible with the planar-composition and TQFT constructions in [11, Theorem 2 and Section 7] and with the reduced complex in [12, Section 3].

A circle-free matching on $2r$ points has r arcs. Gluing it to the r closure arcs produces c circles with $1 \leq c \leq r$. Hence

$$\dim_{\mathbb{F}} R_\Gamma(S) = 2^{c-1} \leq 2^{r-1}.$$

It follows that the total vector-space dimension of $R_\Gamma(C)$ is at most $2^{r-1}M(C)$. Total homology dimension is at most total chain dimension, and homotopy equivalence identifies its homology with $\widetilde{Kh}(\Gamma(P); \mathbb{F})$.

Finally, the graded Euler characteristic of reduced Khovanov homology is the normalized Jones polynomial. Evaluating its grading variable at i corresponds to evaluating the conventional Jones variable at -1 ; all grading monomials have absolute value one. The triangle inequality gives

$$\dim_{\mathbb{F}} \widetilde{Kh}(L; \mathbb{F}) \geq |V_L(-1)| = |\Delta_L(-1)| = \det(L).$$

This is the elementary reduced-homology inequality recorded in [12, Section 3]. The same Euler-characteristic argument works over every field, since the integer graded Euler characteristic of the finite complex is independent of the field. Combining the inequalities proves the result. $\square$

Remark 3.2 (An independent unreduced proof over $\mathbb{F}_2$). The four-ended bound also follows without constructing a reduced closure functor. An ordinary numerator closure sends each circle-free matching to at most two circles and thus at most four scalar generators. Over $\mathbb{F}_2$, the unreduced Khovanov homology is the direct sum of two grading shifts of the reduced homology [13, Corollary 3.2.C]. Therefore

$$4M(C) \geq \dim Kh(N(P); \mathbb{F}_2) = 2 \dim \widetilde{Kh}(N(P); \mathbb{F}_2) \geq 2 \det(N(P)).$$

This supplies the same lower bound $M(C) \geq \det(N(P))/2$.

3.2 Pell growth at a four-ended unknot checkpoint

Use the matrices and tangle conventions of (2.1). For $k \geq 0$, set

$$G_k = (AB)^k, \quad \begin{pmatrix} p_k \\ q_k \end{pmatrix} = G_k \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} r_k \\ s_k \end{pmatrix} = G_k \begin{pmatrix} 3 \\ 2 \end{pmatrix},$$

and choose the standard rational tangle diagrams

$$P_k = T(p_k/q_k), \quad Q_k = -T(r_k/s_k), \quad K_k = N(P_k + Q_k).$$

Theorem 3.3 (Explicit unknots with an exponential intrinsic checkpoint). *There are diagrams K_k of the unknot with $N_k = 8k + 4$ displayed crossings and a specified four-ended prefix P_k such that every explicitly delooped complex C_k retaining the complete Bar–Natan homotopy type of P_k has*

$$M(C_k) \geq \frac{p_k}{2} \geq \frac{1}{2}(3 + 2\sqrt{2})^k = \frac{1}{2}(3 + 2\sqrt{2})^{(N_k-4)/8}.$$

Here

$$p_0 = 1, \quad p_1 = 7, \quad p_{k+2} = 6p_{k+1} - p_k,$$

and, with $\lambda = 3 + 2\sqrt{2}$,

$$p_k = \frac{1 + \sqrt{2}}{2}\lambda^k + \frac{1 - \sqrt{2}}{2}\lambda^{-k}.$$

In particular, the lower bound is exponential in the size of these displayed unknot diagrams, although the rational arithmetic summaries have $O(k)$ bits and decide their closure in $O(k^2)$ bit operations.

Proof. Both columns are primitive and

$$p_k s_k - q_k r_k = \det(G_k) \det \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix} = -1.$$

The rational gluing criterion (2.2) therefore proves that K_k is an unknot. The relevant classical criterion is [7, Theorem 5].

The prescribed construction uses $4k + 1$ crossings for P_k and $4k + 3$ for Q_k , and numerator caps introduce none. For an explicit continued-fraction realization, $P_0 = [1]$ and, for $k \geq 1$,

$$P_k = [\underbrace{2, \dots, 2}_{2k-1}, 3], \quad -Q_k = [\underbrace{2, \dots, 2}_{2k}, 1, 2].$$

The sums of the absolute coefficients are exactly $4k + 1$ and $4k + 3$. These are counts in the displayed input diagrams, not minimal crossing numbers of their knot type.

The matrix

$$AB = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$$

has trace six and determinant one. Its characteristic equation gives the asserted recurrence. Its eigenvalues are λ and λ^{-1} , and the initial values give the stated closed formula. Since

$$p_k - \lambda^k = \frac{\sqrt{2}-1}{2}(\lambda^k - \lambda^{-k}) \geq 0,$$

we have $p_k \geq \lambda^k$.

Moreover, $G_k \equiv I \pmod{2}$, so p_k and q_k are odd. Thus $N(P_k)$ is a rational knot and its determinant is p_k . The determinant assertion is the usual primitive fraction formula for rational numerator closure; it can also be computed from its lens-space double branched cover, whose first homology has order p_k . The tangle fraction and numerator/denominator determinant interpretation are reviewed in [8, Section 4.1, Remark 2]. Apply Lemma 3.1 with $r = 2$ to obtain $M(C_k) \geq p_k/2$. Since $M(C_k)$ is integral and p_k is odd, the rounded bound is $(p_k + 1)/2$.

Each matrix update uses additions and doubling of $O(k)$ -bit integers. The two slope columns and their determinant consequently require $O(k)$ storage bits and $O(k^2)$ schoolbook bit operations, as in Theorem 2.5. $\square$

Corollary 3.4 (Stronger bound when the quantum grading is retained). *For the ordinary graded Bar–Natan complex, the conclusion of Theorem 3.3 strengthens to*

$$M(C_k) \geq p_k + q_k, \quad q_k = \frac{\lambda^k + \lambda^{-k}}{2}.$$

Indeed the two individual circle-free matching types must occur at least p_k and q_k times, respectively, counting all shifts and chain degrees. More generally, for any number of ends a graded version of Lemma 3.1 gives $M(C) \geq \det(\Gamma(P))$.

Proof. Use z for the quantum grading variable to distinguish it from the fraction denominator q_k . A matching whose closure has c circles contributes, up to a grading monomial,

$$(z + z^{-1})^{c-1}$$

to the graded dimension of the reduced closure. At $z = i$ this is zero if $c > 1$ and has absolute value one if $c = 1$. The reduced graded Euler characteristic therefore yields

$$\det(\Gamma(P)) \leq \#\{\text{summands whose } \Gamma\text{-closure is one circle}\}.$$

This proves the general assertion. It only uses graded homotopy equivalence and evaluation of the graded Euler characteristic; it does not assume thinness of Khovanov homology.

For four ends there are exactly two circle-free matchings, the horizontal and vertical ones. Numerator closure gives two circles on the horizontal matching and one on the vertical matching; denominator closure interchanges these roles. Let M_H and M_V be their total multiplicities. The numerator estimate gives $M_V \geq p_k$. The denominator closure of P_k is also a rational knot because q_k is odd, and has determinant q_k . Applying the same estimate to denominator closure gives $M_H \geq q_k$. Thus $M(C_k) = M_H + M_V \geq p_k + q_k$.

The recurrence for q_k is the same trace-six recurrence as for p_k , with initial values $q_0 = 1$ and $q_1 = 3$, giving the displayed formula. One orientation of the prefix need not extend to both test closures. Use the orientation-free bracket complex first; changing strand orientations changes the normalized complex only by global homological and quantum shifts. Their evaluations have absolute value one. Thus both estimates apply to the same matching multiplicities, despite independent closure orientations. $\square$

Scope of the obstruction. This theorem applies at the specified prefix checkpoint and to representations preserving its full homotopy type in the stated explicit basis. It is not a lower bound for every scan order or every unknot-recognition algorithm. An algorithm can use information about the actual continuation, absorb parts of it earlier, or retain the rational slope in a compact arithmetic form. The conclusion also does not charge an exponentially repeated object encoded by a binary multiplicity as exponentially many stored words: it rules out a small *expanded* full complex, and any use of compressed multiplicities or other compressed objects must prove that all required operations on that representation remain efficient.

The distinction from the Boolean identity-matrix theorem is precise. That theorem rules out a small exact linear representation of all Boolean continuation answers on its family. The present theorem rules out a small expanded full homotopy representative for one sequence of prefixes inside unknots. Both leave room for the rank-two signed arithmetic pairing followed by a nonlinear test.

4 Exact arithmetic recognition and local obstructions from Montesinos tangles

This section provides a complete arithmetic decision procedure on an explicitly specified family, and a certificate that can reject an arbitrary diagram containing a suitable local tangle. The topology underlying the complete Montesinos classification is classical. The local unknotability criterion is due to Nogueira and Salgueiro [9, Theorem 4.9]. Our purpose is to state the input model, prove the arithmetic and diagram verification obligations, and make these results available to the recognizer. No statement here asserts efficient discovery of a Montesinos presentation for an arbitrary knot.

4.1 Conventions and exact component counts

Write $R(p/q)$ for the rational tangle of fraction p/q , with boundary points NW, NE, SE, SW in their cyclic order. Horizontal addition joins the east pair of one tangle to the west pair of the next. The numerator closure $N(T)$ joins NW to NE and SW to SE outside the tangle disk. We use the standard fraction rules $R(x) + [a] = R(x + a)$ and $R(x)^{-1} = R(1/x)$; the latter operation is a quarter rotation followed by crossing reversal. A rotation alone sends x to $-1/x$. These conventions, including the distinction between rotation and inversion, are those of [5, Section 2]. An implementation must calibrate its crossing convention against these rules.

The structured input is

$$M(e; p_1/q_1, \dots, p_s/q_s) = N([e] + R(p_1/q_1) + \dots + R(p_s/q_s)),$$

where e and the rational-tangle continued-fraction coefficients are signed binary integers. Every final slope is finite. Each pair is reduced, with $q_i > 0$. A denominator-one summand is absorbed into e . We thereafter write the remaining slopes as β_i/α_i , $1 \leq i \leq r$, where $\alpha_i \geq 2$ and $\gcd(\alpha_i, \beta_i) = 1$. It is optional to impose $0 < \beta_i < \alpha_i$: replacing β_i by $\beta_i - a\alpha_i$ and replacing e by $e + a$ preserves the Montesinos link. This is the standard integer-transfer isotopy; it follows by moving integral twists through rational summands using the rational flip isotopies [5, Section 2]. The tests below do not require the optional remainder normalization. The delivered implementation always performs it.

For a finite continued fraction $[a_1, \dots, a_\ell]$, the product

$$\begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \dots \begin{pmatrix} a_\ell & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$$

computes its projective fraction without floating-point arithmetic. Intermediate infinite values are harmless in this formulation. A final $q = 0$ is outside the stated input model and must produce an unsupported-input response, not a knot verdict. In particular, one must not replace an infinite tangle by an integer summand or apply the formulas below to it.

Lemma 4.1 (Endpoint parity and component count). *Let k be the number of even denominators among the reduced finite slopes in a horizontal Montesinos sum. The sum has $\max(k-1, 0)$ closed strand components in its interior. Consequently it is a genuine two-string tangle with no closed components if and only if $k \leq 1$. Its numerator closure has*

$$c = \begin{cases} k, & k > 0, \\ 1, & k = 0 \text{ and } e + \sum_i \beta_i \text{ is odd,} \\ 2, & k = 0 \text{ and } e + \sum_i \beta_i \text{ is even.} \end{cases}$$

The same formula includes $r = 0$.

Proof. The endpoint pairing of a rational tangle depends only on the coprime pair (p, q) modulo two. There are three possibilities:

$(p \bmod 2, q \bmod 2)$	pairing	type
$(0, 1)$	$NW \leftrightarrow NE, SW \leftrightarrow SE$	H
$(1, 0)$	$NW \leftrightarrow SW, NE \leftrightarrow SE$	V
$(1, 1)$	$NW \leftrightarrow SE, SW \leftrightarrow NE$	X .

This can be verified inductively from $[0]$, since adding one integral twist and inverting the tangle act on the three pairings exactly as they act on (p, q) modulo two. Thus it holds for every continued-fraction construction, including constructions with negative and zero coefficients.

An H or X piece carries the left endpoint pair to the right endpoint pair by the identity or the transposition. If there is no V piece, the total transposition parity is $e + \sum_i \beta_i$; numerator closure gives the cycles of this two-element permutation. If there are $k > 0$ pieces of type V , each interval between consecutive such pieces closes one component. There are $k - 1$ such intervals in the open sum. The two remaining outer caps close one additional component under numerator closure, giving k components. No crossings are removed or changed in this argument: it follows the strands through the actual tangle connectivity. $\square$

The component check is mandatory. A link with several components is outside the ordinary knot-recognition domain; its presence is not evidence for a nontrivial knot. In a local witness the corresponding mandatory check is that the selected portion consists of two open strands and no internal closed strand. Following opposite darts through the crossings verifies this directly, independently of the arithmetic parity computation.

4.2 A direct nontriviality certificate for the branched cover

The branched double cover of a Montesinos link is Seifert fibered, with a presentation, in the above sign convention,

$$\pi_1 \Sigma_2(M) = \langle h, x_1, \dots, x_r \mid [h, x_i] = 1, x_i^{\alpha_i} h^{\beta_i} = 1, x_1 \cdots x_r = h^e \rangle. \quad (4.1)$$

This is the classical Montesinos construction; see the application and attribution in [9, proof of Theorem 4.9]. Here a lift of each rational-tangle ball is a solid torus. The central piece is the circle bundle over the punctured sphere, and the meridian of the i th attached solid torus is $\alpha_i x_i + \beta_i h$. These gluing slopes give the displayed relations by van Kampen's theorem.

We give an explicit elementary representation to avoid treating the nontriviality of a triangle group as a group-decision black box.

Lemma 4.2 (An explicit nontrivial triangle representation). *For all integers $a, b, c \geq 2$, the group*

$$\langle x, y, z \mid x^a = y^b = z^c = xyz = 1 \rangle$$

has a nontrivial representation in $\mathrm{PSL}_2(\mathbb{C})$.

Proof. Set

$$\lambda = e^{\pi i/a}, \quad t = 2 \cos(\pi/b), \quad s = 2 \cos(\pi/c), \quad d = \lambda - \lambda^{-1},$$

and define

$$u = \frac{s - \lambda^{-1}t}{d}, \quad v = \frac{\lambda t - s}{d}, \quad X = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}, \quad Y = \begin{pmatrix} u & 1 \\ uv - 1 & v \end{pmatrix}.$$

The denominator d is nonzero because $a \geq 2$. Direct calculation gives $\det X = \det Y = 1$, $\text{tr } Y = t$, and $\text{tr}(XY) = s$. Thus the distinct eigenvalues of Y are $e^{\pi i/b}$ and $e^{-\pi i/b}$, while those of XY are $e^{\pi i/c}$ and $e^{-\pi i/c}$. It follows that

$$X^a = Y^b = (XY)^c = -I.$$

Sending x, y, z to the projective classes of $X, Y, (XY)^{-1}$ therefore satisfies every relation. The image of x has order exactly a , so it is nontrivial. $\square$

This is a uniform symbolic certificate formula, rather than a requirement to expand cyclotomic fields of possibly enormous degree. Its instantiated parameters are just the three binary integers a, b, c . The verifier checks that they are the selected denominators and that they are at least two; the lemma supplies the representation for every such triple. No numerical cosine evaluation or unproved finite-group search is involved.

Proposition 4.3 (Three genuine exceptional fibers obstruct the unknot). *If $r \geq 3$, a Montesinos link of the stated form cannot be the unknot.*

Proof. In (4.1), quotient first by $h = 1$, then by $x_i = 1$ for all but any three selected indices, retaining their cyclic order. The resulting quotient is the triangle group with orders equal to their denominators. It is nontrivial by Lemma 4.2. Hence the branched double cover is not simply connected. The double cover branched over the unknot is S^3 , whose fundamental group is trivial. $\square$

4.3 A complete decision procedure on the structured input

Put

$$Q = \prod_{i=1}^r \alpha_i, \quad D = eQ + \sum_{i=1}^r \beta_i \frac{Q}{\alpha_i}, \quad (4.2)$$

with the empty-product convention $Q = 1$. Abelianizing (4.1) gives a square integer relation matrix whose determinant has absolute value $|D|$. Indeed, eliminate the diagonal block $\text{diag}(\alpha_1, \dots, \alpha_r)$ over $\mathbb{Q}$; its Schur complement is $-e - \sum_i \beta_i / \alpha_i$. When $D \neq 0$, the order of first homology is $|D|$; when $D = 0$, first homology is infinite.

Theorem 4.4 (Complete Montesinos recognition in compressed arithmetic). *For a structured finite-slope Montesinos input, first compute the component count in Lemma 4.1. If it is one, the represented knot is the unknot exactly when $r \leq 2$ and $|D| = 1$.*

Proof. The case $r \geq 3$ is Proposition 4.3. For $r = 0$ or $r = 1$, the link is the numerator of a rational tangle. For $r = 2$, absorb the integral summand into one rational tangle; the numerator closure of the sum of two rational tangles is a two-bridge link. This last statement concerns the closure, and does not assert that the open sum is rational; see [5, Section 2]. The branched cover of a two-bridge link is a lens space, and its numerator parameter has absolute value $|D|$. The rational-link classification says that a two-bridge knot is the unknot precisely when the absolute numerator is one [6]. This proves the claimed equivalence. Equivalently, the $r = 2$ assertion is the integral-closure calculation in [9, Theorem 4.9]. $\square$

Proposition 4.5 (Bit complexity of the arithmetic implementation). *Let B be the complete binary length of a structured description, including coefficients, signs, and list delimiters. The production arithmetic classifier uses $O(B^2)$ schoolbook bit operations and $O(B)$ bits of integer data. Its valid branch certificate has $O(B)$ bits. The independent verifier in the package has the conservative bound $O(B^3)$ on these certificates. Input parsing must additionally charge the*

actual size of a purported certificate. These are bounds in compressed source length; they do not include PD expansion.

Proof. Give a coefficient a the weight $b(a) = 1 + \lceil \log_2(1 + |a|) \rceil$ and charge at least one bit per delimiter. The sum of all weights is $O(B)$. The column recurrence starts at $(p, q) = (1, 0)$ and, reading a continued fraction backwards, replaces it by $(ap + q, p)$. Every matrix has determinant -1 , so the final pair is primitive without a gcd computation. The one-norm of a column is bounded by the product of $(1 + |a|)$ over its coefficients; allowing a constant for the initial state gives $O(B)$ bits for every entry, even with zeros and signs. The total size of the final numerator–denominator pairs over all input summands is $O(B)$ as well.

More precisely, a recurrence with current entry length s and coefficient weight b costs $O(sb + s)$ with schoolbook arithmetic. Summing these costs is $O(B^2)$, since all partial lengths are bounded by the corresponding partial sum of coefficient weights and there are $O(B)$ entries. Changing the sign to make $q > 0$ is linear in its size. For each finite pair, Euclidean division $p = aq + \beta$ computes $0 \leq \beta < q$. The sum of the squares of the pair lengths is at most the square of their total length, so all these divisions together still cost $O(B^2)$. The implementation absorbs every integral part into e and drops exactly the pairs with $q = 1$.

To evaluate (4.2), begin with $(d, Q) = (e, 1)$ and, for each remaining (β, α) , update

$$(d, Q) \mapsto (d\alpha + \beta Q, Q\alpha).$$

No rational cancellation is performed. The total bit lengths of the (β, α) pairs are $O(B)$, and every accumulated entry has $O(B)$ bits. Weighting each multiplication by both operand lengths shows that the sum of multiplication costs is $O(B^2)$; the at most $O(B)$ additions and parity checks contribute the same bound. All stored slopes, accumulated integers, counts, and the certificate together use $O(B)$ bits of integer and encoded source data. This is an arithmetic-space statement, not a measurement of Python object overhead or process memory.

The production code tracks the maximum bit length of its recorded continued-fraction and accumulator states with a constant number of integer operations per state, costing $O(B^2)$ bit operations in total. The independent verifier reconstructs each fraction by a matrix product and recomputes product–sum expressions for the certificate. Its current replay of the prefix statistics contains nested loops. Bounding their work by $O(B^3)$ is safe: final products have $O(B)$ bits and the sums of operand lengths are $O(B)$. It never needs to expand a cyclotomic field, factor an integer, or enumerate a finite quotient. For valid certificates all compared fields have total length $O(B)$. $\square$

The statistic `max_entry_bits` means the maximum *tracked* continued-fraction or accumulator entry length. It is not the maximum length of every multiplication temporary, and is not peak memory. Temporary products still have $O(B)$ bits by the same estimates. For instance, the calculation for $M(-2; 7/9)$ can form the temporary -18 while the tracked final numerator is -11 ; the former has one more binary digit. The theorem does not infer a space bound from this empirical statistic.

The one-component check can be implemented especially simply as D odd. If exactly one denominator is even, its term $\beta_i Q / \alpha_i$ is odd and every other term in (4.2) is even. If at least two are even, every term is even. If none is even, the parity is $e + \sum \beta_i$. Lemma 4.1 therefore shows that D is odd if and only if the numerator closure has one component. This equivalence includes zero exceptional fibers, and justifies the code’s exact component rejection without expanding a diagram.

4.4 An Alexander-trivial family of genus-one knots

For odd integers p, q, r , use the conventional pretzel diagram $P(p, q, r) = M(0; 1/p, 1/q, 1/r)$, with signs of the twist boxes as displayed in [10, Figure 1 and Theorem 1(3)]. Its usual two-disk, three-band spanning surface is orientable: all three bands have odd twisting, so the disk orientations can be chosen compatibly. It has Euler characteristic $2 - 3 = -1$ and one boundary component, hence genus one. Take one oriented cycle through the first and second bands and another through the second and third bands, choosing their intersection to be $+1$. In this basis a Seifert matrix is

$$V = \frac{1}{2} \begin{pmatrix} p+q & q+1 \\ q-1 & q+r \end{pmatrix}. \quad (4.3)$$

For completeness, the self-pairings are half the signed twists in the two bands traversed by the corresponding cycle. In the shared middle band the off-diagonal pairings are $(q+1)/2$ and $(q-1)/2$: the parallel push-off on the two sides of the single transverse disk intersection differs by one. Their difference is the chosen intersection number, as independently required by $V - V^T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Reversing the entire diagram convention replaces the corresponding form by its mirror form and does not change the conclusions below. There is also an independent check from the skein-derived formula in [10, Theorem 1(3)].

Writing

$$A = \det V = \frac{pq + pr + qr + 1}{4},$$

direct expansion gives

$$\det(V - tV^T) = A(t-1)^2 + t, \quad \Delta_{P(p,q,r)}(t) = 1 + A(t + t^{-1} - 2).$$

Thus $pq + pr + qr = -1$ gives Alexander polynomial one, determinant one, and signature zero: the determinant of $V + V^T$ is $4A - 1 = -1$, so its two eigenvalues have opposite signs. The characterization by this equation for three odd parameters is established mathematics [10, Section 2.3].

In particular, for every odd $a \geq 3$ let

$$p = -a, \quad q = a + 2, \quad r = \frac{a^2 + 2a - 1}{2}. \quad (4.4)$$

All three parameters are odd and have absolute value at least three. Also $p + q = 2$ and $pq = -a(a+2)$, whence $pq + pr + qr = -a(a+2) + 2r = -1$. The input is a knot by Lemma 4.1; its three genuine denominators establish nontriviality by Proposition 4.3. Consequently its Seifert genus is exactly one, although both the Alexander polynomial and the signature give the same answers as for the unknot. The first example is $P(-3, 5, 7)$, with $V = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$. This is a useful regression family for the arithmetic branch. With the implementation's positive-denominator normalization, its signed Montesinos integer D is $+1$; the value $pq + pr + qr = -1$ uses signed pretzel parameters. Their absolute values agree, as required. The classical alternative $(-a, 2a-1, 2a+1)$ has the same Alexander-triviality property; neither family is claimed here as a new class of knots.

4.5 Closure-independent forbidden tangles

An essential distinction is that a tangle may occur inside a much more general knot diagram. The appropriate local property is whether the tangle can occur inside *any* unknot, not whether one selected closure is knotted.

Theorem 4.6 (Nogueira–Salgueiro local obstruction, with input checks). *Let*

$$T = [e] + R(\beta_1/\alpha_1) + \cdots + R(\beta_r/\alpha_r), \quad \alpha_i \geq 2, \quad \gcd(\alpha_i, \beta_i) = 1,$$

be a genuine two-string tangle with no internal closed components and $r \geq 2$. Then T embeds in some unknot if and only if $r = 2$ and

$$F := \beta_1\alpha_2 + \beta_2\alpha_1 \equiv 1 \text{ or } -1 \pmod{\alpha_1\alpha_2}. \quad (4.5)$$

Therefore $r \geq 3$, or failure of (4.5) when $r = 2$, is a nontriviality certificate for every knot containing T .

Proof. For $e = 0$ this is [9, Theorem 4.9(a)]; integral twists can be absorbed into a rational summand, changing F by a multiple of $Q = \alpha_1\alpha_2$ and leaving the criterion unchanged. Here is the cover argument identifying the exact obstruction. The cover of the tangle ball has base disk and exceptional orders $\alpha_1, \dots, \alpha_r$. Its boundary is incompressible for $r \geq 2$. For example, its group has the relations $[h, x_i] = 1$ and $x_i^{\alpha_i} h^{\beta_i} = 1$, without the last relation of (4.1). In the quotient by h it is the free product $C_{\alpha_1} * \cdots * C_{\alpha_r}$, and the boundary base loop maps to $(x_1 \cdots x_r)^{-1}$, an element of infinite order by reduced normal form. The fiber h itself has infinite order, since the homomorphism to $(\mathbb{Q}, +)$ given by $h \mapsto 1$ and $x_i \mapsto -\beta_i/\alpha_i$ respects the relations. Thus the two boundary generators are independent and the boundary torus group injects. In particular, the tangle is essential: a disk separating its two strings would lift to a compression disk of that boundary torus.

If an essential two-string tangle occurs in an unknot, its complementary tangle is rational [9, Proposition 4.4]. A finite complementary slope with denominator at least two would add another genuine exceptional fiber; Proposition 4.3 rules this out. The infinite complementary slope is ruled out separately: its closure is the denominator closure of the horizontal sum, whose determinant is $\prod_i \alpha_i > 1$. This product also follows by expressing the denominator closure as the connected sum of the denominator closures of the rational pieces. Hence the only possible case is $r = 2$ with an integral complement $[u]$. Its signed homology determinant is

$$F + (e + u)Q.$$

The resulting link is rational and is the unknot precisely when this integer is $\varepsilon \in \{1, -1\}$, giving exactly (4.5). Conversely a successful sign gives the explicit integral unknotting complement $u = (\varepsilon - F)/Q - e$. Since $Q \geq 4$, at most one of the two signs works. $\square$

The genuine-tangle hypothesis can either be checked by following strands or by verifying that at most one denominator is even. It cannot be dropped merely because the syntax contains rational summands. The arithmetic witness for the two-summand rejection branch is simply the recomputed residue $F \bmod Q$, which must be neither 1 nor $Q - 1$. No factorization of Q is needed. Satisfying the congruence gives *no positive verdict for an arbitrary surrounding knot*: it only says that some choice of complement could unknot this tangle.

For example, the tangle $R(1/3) + R(1/4)$ has $F = 7$, $Q = 12$, so it is forbidden in every unknot. Its numerator and denominator determinants are 7 and 12, with gcd one. Thus the obstruction is stronger on this example than the gcd-of-two-closures determinant obstruction: neither local homology order alone supplies the universal obstruction. Likewise the first two boxes of the family (4.4) already give the forbidden tangle $R(-1/a) + R(1/(a+2))$: here $F = -2$ and $Q = a(a+2) \geq 15$. This local witness remains valid if the entire third-box region is replaced by any exterior tangle giving a one-component closure.

4.6 A rigorous planar certificate for occurrence in a PD diagram

We now specify the missing link between arithmetic data and a supplied knot diagram. A list of fractions, or an abstract four-edge cut, is not by itself an occurrence certificate.

Let H be the expanded connected shadow of the tangle constructed by the declared rational-tangle operations. It has four degree-one port vertices; its other vertices are crossings with a cyclic order and an overpassing opposite pair. Let G be the validated connected plane shadow of the input knot. A certificate maps the crossing darts of H injectively to the crossing darts of G . It must satisfy all the following requirements:

1. Crossing images are distinct; each crossing map preserves its cyclic order and the overpassing pair. The same global orientation convention is used throughout.
2. Every internal edge pairing of H equals the corresponding edge pairing in G . Every dart at an image crossing is accounted for, either by an internal edge or by one of the four ports. There are no unrecorded attachments or ignored incident darts.
3. Each of the four port darts belongs to a distinct edge of G whose other endpoint is outside the selected crossing set. None of these edges is an internal edge of the pattern. This deliberately strict condition is the initial implementation contract; allowing two cut points on one exterior edge is a possible extension, not an accepted certificate here.
4. In the plane ribbon graph H , all four ports lie on one facial boundary, in the declared boundary order. This is checked by the face permutation of the rotation system after adding the four degree-one vertices.
5. Following opposite darts through all crossings of H gives exactly two open strands and no closed strand component.

The builder's rational fractions are recomputed from its operations. Matching only fractions, crossing counts, or abstract unembedded graphs does not check these requirements.

Lemma 4.7 (The common-face certificate yields a separating disk). *Under the preceding conditions there is a disk in the projection sphere whose interior contains precisely the certified tangle diagram and whose boundary meets the knot in its four declared ports. In particular, the input knot contains the declared tangle, up to the boundary-preserving diagram isotopy.*

Proof. Embed H in the projection sphere using the certified dart map, placing a cut point in each of the four distinct outgoing edges. Take a small regular neighborhood of this connected embedded graph. The boundary component corresponding to the declared face is incident to all four outgoing port segments. Every component of the remaining diagram attaches to H at a port: otherwise it would be a separate component of the connected shadow G . Since no edge crosses H , all such components lie in that one face. The other boundary components of the regular neighborhood therefore enclose no part of the remaining diagram. Fill those other boundary components with their complementary disks. This produces a disk; adjust its remaining boundary near the four terminal port segments so that it meets each once and no other diagram edge. Its port order is the certified one. Thickening this planar disk slightly in the projection direction gives the required tangle ball in S^3 . $\square$

Proposition 4.8 (Verification and deterministic pattern search). *For a fixed connected tangle pattern with m crossings and an input plane knot diagram with n crossings, the occurrence certificate can be verified in $O(n + m)$ word operations after parsing. All its orientation-preserving dart occurrences can be found in $O(nm)$ word operations, plus parsing and output costs. A conservative bit bound adds a factor $O(\log(n + m))$.*

Proof. All edge, rotation, strand, injectivity and face checks are traversals of finite permutations and graphs. For the search, choose one crossing of the connected pattern as root. Its image and cyclic offset have exactly $2n$ potential choices: offsets zero and two preserve the overpassing opposite pair. An internal edge then forces the image of the next crossing and the image of its incoming dart; preservation of cyclic order forces its remaining three darts. Propagate this rule through the connected pattern, rejecting inconsistent revisits, collisions, and overpair violations. There are no further choices after the root choice. Each candidate takes $O(m)$ work, including the common-face and port checks. The global validation of G is performed once. Direct indexed arrays give the stated word bound; indices have $O(\log(n + m))$ bits. $\square$

For a supplied already compiled template, this is the matching cost. The public implementation additionally validates and relabels the target PD; its sorting step gives $O(n \log(n + 2) + nm)$ word operations with bounded labels. Source reading, arithmetic, and template compilation have their own B - and m -dependent cost and must be added if a template was not already compiled. In particular, an arbitrarily long source with zero coefficients is not processed in time bounded by m alone.

The pattern expansion cost is explicit here. If the chosen literal continued-fraction pattern has more than n crossings, it cannot have such an injective occurrence in the supplied PD and may be discarded before expansion. This does not affect the independent compressed-input theorem, which never expands twist exponents. Searching a library of patterns with total expanded size M costs $O(nM)$ by the same argument. Efficient recognition of all Montesinos tangles hidden behind arbitrary diagram isotopies is a different, unsolved implementation obligation in this package.

In particular, a seven-crossing template for $R(1/3) + R(1/4)$ gives an $O(n \log(n + 2))$ -time public rejection stage (with a linear matching core) on every diagram containing an admissible occurrence with four distinct outgoing cut edges. This is an explicit family with unrestricted exterior diagram size and complexity. It is not a statement that every nontrivial knot has such a tangle, or that finding it after arbitrary isotopies has linear complexity.

The useful integration direction is one-sided: a verified occurrence plus Theorem 4.6 gives a rigorous KNOTTED verdict. Failure to find a pattern, failure of a resource budget, or arithmetic that allows an unknotting closure causes the normal pipeline to continue.

4.7 Bounded-description discovery without a supplied sidecar

The preceding matching procedure also gives an explicit discovery bound when the local tangle has a short literal description. This removes the need for a user-supplied occurrence certificate on a precisely delimited class. It does not require a conjecture about the descriptions of all nontrivial knots.

For integers $n \geq 1$ and $h \geq 2$, let $\mathcal{S}(n, h)$ consist of all ordered lists of at least two nonempty continued-fraction blocks

$$s = ([a_{1,1}, \dots, a_{1,\ell_1}], \dots, [a_{b,1}, \dots, a_{b,\ell_b}]), \quad b \geq 2,$$

such that

$$\sum_j \ell_j \leq h, \quad -n \leq a_{j,k} \leq n, \quad \sum_{j,k} |a_{j,k}| \leq n.$$

Its specified diagram is obtained by the fixed rational-tangle compiler and horizontal composition, with initial integer term $e = 0$. No diagram isotopies are performed during matching. The compiler expands an integer a into $|a|$ crossings and implements tangle rotation and reflection without adding crossings. Thus its literal pattern has at most n crossings. Zero

coefficients and repeated descriptions are allowed; they only make the following upper bound more generous.

Call a source *admissible and forbidden* when every original final fraction is finite and nonintegral with denominator greater than one, its compiled shadow is connected, all four ports pass the common-face test, its strand tracing gives exactly two open strands and no closed component, and Theorem 4.6 rules out every unknotting complement. Occurrences must have four distinct external cut edges, as above. This matches the strict source grammar of the delivered local compiler: initial $e = 0$, at least two nonintegral rational summands, and no discarded integral summands. Integer parts within a nonintegral slope do not affect its congruence obstruction. The graph being matched is always the original literal compiled graph.

Proposition 4.9 (Deterministic discovery from bounded source length). *Given a validated n -crossing plane knot diagram G and an integer $h \geq 2$, there is a deterministic one-sided search with running time*

$$(n + 2)^{O(h)}$$

*in bit operations which returns **KNOTTED** if and only if G contains an orientation-preserving dart occurrence, satisfying Lemma 4.7, of a pattern compiled from some admissible and forbidden source in $\mathcal{S}(n, h)$. When it returns **KNOTTED**, it also returns the source and the verified crossing-dart map. Otherwise it returns **UNKNOWN**. In particular, for $h \leq C \log(n + 2)$ with fixed C , this restricted witness-discovery problem has running time*

$$\exp(O((\log(n + 2))^2)) = (n + 2)^{O(\log(n+2))}.$$

Proof. A source with total coefficient count ℓ is specified by a sequence of ℓ coefficients and by selecting separators among the $\ell - 1$ gaps. There are $(2n + 1)^\ell$ sequences and at most $2^{\ell-1}$ separator sets; requiring at least two blocks and imposing the expansion budget can only decrease their number. Consequently

$$|\mathcal{S}(n, h)| \leq \sum_{\ell=1}^h (2n + 1)^\ell 2^{\ell-1} \leq (2(2n + 1))^{h+1}.$$

Enumerate these finite sequences and separators, and immediately discard any source exceeding the expansion budget. Evaluate every fraction by its integer matrix product, reject integral final summands, and apply the exact component and local obstruction tests. The source bit length is $B = O(h \log(n + 2))$; all these arithmetic operations cost $B^{O(1)}$ by the previous estimates. Unsupported final infinite fractions are discarded.

Compile each remaining source, checking connectedness, its ports, and strand components. Standard explicit tangle composition and index permutations are polynomial in the source and expanded lengths. Even a compiler which revisits the current graph at every continued-fraction operation takes $O(hn)$ word operations, absorbed by $O(n^2 + B^2)$ up to logarithmic index costs. Discard any inadmissible result. For an admissible forbidden pattern with $m \leq n$ crossings, run the root-dart propagation of Proposition 4.8, costing $O(nm) \leq O(n^2)$ word operations. Including integer arithmetic and bit indices, with no fixed implementation expansion cap, the work per enumerated source is bounded by

$$O(n^2 \log(n + 2) + B^{C_0})$$

for some fixed constant C_0 . Multiplication by the preceding source count is $(n + 2)^{O(h)}$; polynomial factors in h are absorbed since $h \geq 2$.

Any returned map passes the separating-disk verification and the exact local obstruction theorem, so its verdict is sound for the surrounding knot. Conversely, every source in the stated

finite class is enumerated. If its literal pattern has such an occurrence, the matching procedure tries the correct root image and cyclic offset and then reconstructs the entire map. It therefore finds that occurrence. Substituting $h \leq C \log(n + 2)$ gives the displayed quasi-polynomial bound. $\square$

This is a theorem about discovering local certificates of bounded literal description length in the supplied diagram. Its completeness class is specified by the source grammar and the checked plane embedding, rather than by knot type alone. A diagram requiring Reidemeister moves before such a pattern becomes visible need not be rejected by the search. No bound on h for arbitrary nontrivial knot diagrams is proved or assumed. The seven-crossing fixed-pattern matcher is an immediately usable implementation of the underlying search primitive; the exhaustive source enumerator in Proposition 4.9 is a separate algorithmic construction, and should be distinguished from that implementation in benchmark claims. The code defaults to a 100,000-crossing expansion cap; that resource policy is not part of the abstract completeness theorem. A fixed finite pattern library and a bounded exhaustive enumerator may both precede the existing exact fallback recognizer.

5 Implementation, certificates, and integration contracts

The package includes a runnable copy of the updated **fast** recognizer and a patch against the pinned baseline. The patch changes or adds thirteen paths, including two new mathematical modules, their tests, source-provenance handling, CLI integration, documentation, and local certificate fixtures. The existing license is preserved. This section describes the public behavior that a repository integration should retain.

5.1 Structured arithmetic and actual diagram construction

The arithmetic API accepts an integer and a sequence of finite continued fractions:

```
from fastunknot.rational import (
    montesinos_certificate, verify_montesinos_certificate,
)

source = [[0, -3], [0, 5], [0, 7]]
certificate = montesinos_certificate(0, source)
assert certificate["status"] == "KNOTTED"
assert certificate["determinant"] == 1
assert verify_montesinos_certificate(0, source, certificate)
```

This call performs no PD expansion. It is the compressed-input interface to Theorem 4.4. In particular, native Python integers with many thousands of binary digits are supported by the arithmetic path. The bit model in Proposition 4.5 describes those integers, rather than a unit-cost model for arbitrarily large numbers.

For integration into the ordinary pipeline, the diagram constructor expands the same source into a real plane diagram:

```
from fastunknot import Diagram, recognize

d = Diagram.from_rational(0, [[0, -3], [0, 5], [0, 7]])
assert d.crossings == 15
answer = recognize(d)
assert answer.status == "KNOTTED"
```

```
assert answer.method == "montesinos-orbifold-quotient"

# Alias with the mathematical family name:
same = Diagram.from_montesinos(0, [[0, -3], [0, 5], [0, 7]])
```

The compiler implements horizontal twist composition, quarter rotation, crossing reversal, and numerator closure. Rotation changes a slope to $-1/x$; reciprocal construction adds crossing reversal to obtain $1/x$. A lazy crossing-switch flag prevents repeated physical mirroring of all already constructed rows at every reciprocal step. Its final output still undergoes the ordinary spherical-embedding and component checks.

The corresponding JSON representation is:

```
{
  "montesinos": {
    "e": 0,
    "tangles": [[0, -3], [0, 5], [0, 7]]
  }
}
```

The CLI accepts this through the ordinary input loader. The switch `--no-rational` disables the new early decision, which is useful for paired experiments on the very same expanded diagram. This switch does not change what knot the supplied source represents.

The arithmetic stage recognizes both signs of the determinant-one two-bridge condition. For instance, `montesinos_certificate(0, [[1], [-1, -2]])` describes $N(T(1) - T(3/2))$ and returns `UNKNOT`. For $P(-3, 5, 7)$, in contrast, the three exceptional denominators establish nontriviality even though the determinant is one. The latter decision is not an inference from the Alexander polynomial.

5.2 Provenance is part of the correctness argument

An arbitrary PD accompanied by an unrelated list of fractions is not covered by the structured theorem. The implementation therefore attaches immutable, detached rational source data only when its own checked constructor creates the PD. The ordinary `Diagram.from_pd` constructor does not invent or infer this provenance. Mutating the caller's original list after construction does not mutate the stored source.

Mirroring preserves the connection between data and topology by changing the source consistently. Pickle roundtrips replay construction and verify the resulting source/PD relation. A guard also prevents a second `Diagram.__init__` call from replacing the PD of an already initialized object while leaving old provenance attached. These are ordinary supported API invariants; the mathematics does not assume an attacker can rewrite arbitrary internal interpreter objects.

This contract allows `recognize` to use a source without solving a second, potentially difficult tangle-recognition problem. When the source is absent, the existing PD pipeline is available. Automatic extraction of rational structure from general diagrams is an explicit next research task.

5.3 Input rejection and resource behavior

The following cases are handled before a topological verdict is returned. Every coefficient must be a literal integer; Python booleans are rejected despite their being integer subclasses.

Continued-fraction blocks must be nonempty. Intermediate projective infinity is allowed, but a final denominator zero is outside the supported source grammar. A final multi-component numerator closure is outside the knot domain and is rejected as input, rather than labeled a nontrivial knot.

The PD constructor defaults to a literal expansion ceiling of 100,000 crossings, checked before allocating the diagram. This policy protects the expanded interface from exponentially large C . It is separate from the compressed arithmetic API, which does not need to allocate C crossings. The current CLI uses the expanded constructor; the very large compressed benchmarks below use the direct arithmetic API.

The existing recognition timeout and complex-object ceilings retain their original meaning. Exhausting those limits yields UNKNOWN, never UNKNOT. The separate expansion ceiling instead rejects construction with a `DiagramError`; the CLI reports invalid input. Construction precedes the recognition timer and is not covered by that timer. The exact arithmetic branch is still subject to ordinary input processing and the chosen pipeline budget; it does not create a universal real-time guarantee.

5.4 Local occurrence certificates in arbitrary PDs

The second new module is `fastunknot.tangle_obstruction`. Its direct verification interface accepts a host diagram and a source-plus-map certificate:

```
from fastunknot.tangle_obstruction import (
    find_subtangle_obstruction,
    verify_subtangle_certificate,
    recognize_with_subtangles,
)

certificate, evidence, stats = find_subtangle_obstruction(host)
if certificate is not None:
    checked = verify_subtangle_certificate(host, certificate)
    assert checked["status"] == "KNOTTED"

# Optional search, followed by the existing recognizer if no witness is found.
answer = recognize_with_subtangles(host, seconds=3.0)
```

Here `host` is a normal validated `Diagram`; it need not carry rational provenance. The example fixture files in `fast/examples` are plain PDs, and their local certificates are separate files in `fast/certificates`.

Each certificate contains exactly a version, a pattern source, a list of crossing images, and a list of cyclic offsets:

```
{
  "version": 1,
  "pattern": {"e": 0, "tangles": [[0, 3], [0, 3]]},
  "crossings": [...],
  "rotations": [0, 2, ...]
}
```

The ellipses above indicate the format; the delivered certificates contain complete integer lists. Verification compiles the source or retrieves a previously checked immutable template from the compilation cache. It then checks the host and occurrence afresh: the crossing map, the cyclic order, the overpassing pairs, every internal edge, the four distinct outgoing edges, the common

port face, and the two open strands. The checked template supplies the exact fractions and obstruction. A stated knot verdict in a JSON object is never itself a certificate.

The initial compiler accepts only $e = 0$, at least two finite nonintegral rational summands, a connected crossing shadow, two open strands with no closed component, and four distinct external cut edges at an occurrence. The full arithmetic classifier has a broader source grammar; these two interfaces should not be conflated.

The default catalogue consists of six templates: the sums with slopes $(1/3, 1/3)$, $(1/2, 1/5)$, and $(1/3, 1/4)$, and their mirrors. Their literal sizes are six or seven crossings. Their respective uncanceled congruence numerators are 6, 7, 7, with products 9, 10, 12. Each fails the ± 1 congruence and has at most one even denominator.

Connectedness makes matching unusually simple. Once a root crossing image and an allowed half-turn are selected, every internal edge forces its neighbor's image and rotation. The search does not enumerate all n^m crossing maps. For an already compiled m -crossing pattern it uses the $O(nm)$ forced-propagation algorithm proved above. The public call also sorts labels during validation, adding $O(n \log(n + 2))$ word operations, plus source reading and compilation. These deterministic word bounds can be realized with indexed arrays. The Python implementation uses dictionaries and sets in some checks; its operation accounting uses the usual constant-time table model. No worst-case bit bound for adversarial interpreter hash-table behavior is being inferred from a wall-clock measurement.

The wrapper is opt-in. An inconclusive search continues to the existing pipeline, and a failed time budget has the same UNKNOWN interpretation. Successful local rejection is independent of the exterior's knot polynomial or homology. An unknottable local pattern, however, supplies no positive verdict about its actual exterior.

5.5 What the integration patch does and does not contain

The patch includes `rational.py`, `tangle_obstruction.py`, the new tests and fixtures, modifications to the diagram class and recognition entry points, and updated documentation. It can be reviewed and applied from the ProveIt repository root; the exact commands and baseline checks are in `integration/README.md`.

The archive also contains the entire runnable `fast` directory, the article sources and PDF, example-source generators, independent diagnostics, raw measurements, and manifests. The theoretical exhaustive enumerator of Proposition 4.9 is not silently presented as shipping software. The implementation supplies the required fixed-pattern search primitive; a practical enumerator should be a separate reviewed contribution with source deduplication and explicit budgets.

6 Experiments, independent checks, and practical effect

The experiments test mathematical conventions, certificate soundness, integration behavior, and the performance of particular workloads. They do not establish an asymptotic upper bound. All example sources, plain PD fixtures, complete certificates, seeds, commands, and raw timing samples are included in the archive.

6.1 Validation layers

The unchanged pinned baseline passes 134 test methods. The final integrated suite passes 152 methods, including twelve new arithmetic methods and six new local-pattern methods. The final full-suite run took 30.415 seconds as reported by `unittest`, with measured process

wall time 30.651362 seconds and exit code zero. The baseline and integrated logs are kept separately, so an old documentation count does not obscure the actual test coverage.

The new tests exercise 76 small rational closures and 56 small two-tangle closures against an independently constructed scalar Khovanov cube; 160 random Montesinos examples are checked against exact Alexander determinants. They also cover a small complete identity matrix, determinant-one knots with exceptional orders $(2, 3, 5)$ and $(2, 3, 7)$, negative and zero coefficients, intermediate infinity, final-infinity rejection, mirror and pickle behavior, mutation and certificate corruption, expansion ceilings, large integer coefficients, and local occurrence invariance.

Three further diagnostics are independent of the production arithmetic decision:

1. The identity-family script verifies all $\sum_{m=0}^{10} 4^m = 1,398,101$ entries through $m = 10$. The largest matrix is 1024×1024 . It checks primitivity, parity, diagonal determinants, the stronger off-diagonal gap, and a separately implemented twist/rotation program. The topological assertion is supplied by Theorem 2.2, not by these finite checks.
2. The independent Montesinos audit evaluates 1,200 sources: 718 one-component constructed PDs against exact Alexander determinants and 482 link cases against actual component counts. It also performs 1,436 mirror/pickle roundtrip checks and verifies the reinitialization guard.
3. The independent local audit rejects twelve forged certificate variants, invalidates a witness after a selected crossing switch, checks forty combinations of crossing reordering, relabeling, half-turns, and mirroring, and finds no false rejection on 31 known identity-family unknots. It also verifies a three-exceptional local occurrence and rejects patterns that are unknottable, integral, or have an internal closed component.

The checkpoint diagnostic constructs 21 actual diagrams, for $0 \leq k \leq 20$, checks the two continued fractions and exact counts $8k + 4$, and verifies their arithmetic unknot certificates. For the small prefix closures it also performs fourteen independent numerator/denominator Alexander checks and six independent scalar-cube rank checks. It checks the ordinary ungraded and strengthened graded lower-bound formulas separately.

Finally, the integration patch was applied to an isolated exact baseline. All 81 resulting noncache files matched the delivered `fast` snapshot byte for byte. The eighteen new test methods passed there with `PYTHONPATH` removed and no dependence on the outer experiment directory. This checks that the proposed repository integration is self-contained. It does not certify a proof-assistant formalization.

6.2 Timing protocol and comparison scope

The recorded environment is Python 3.12.14 on x86-64 Linux. The benchmark uses a warm in-process interpreter, seven paired rounds per case, a three-second budget, and a 20,000-object complex ceiling. Every timed recognition call receives a freshly constructed and validated diagram, so neither branch benefits from previously populated geometry caches.

For each structured case a round executes the old path A , five samples of the new path B , and A again. The round's baseline is the mean of the two A times; its new time is the median of the five B samples. Reported times are medians across rounds, and reported speedups are medians of the seven paired ratios. Thus the ratio of the two separately reported medians need not equal the reported paired speedup.

The old path is the updated code with `use_rational=False`, on the same expanded PD as the enabled path. It measures the effect of the added stage without changing the topology

or input ordering. Recognition-stage timing excludes construction for both paths. A separate column times new PD construction plus recognition. No claim about CLI startup, disk I/O, or deployment latency is drawn from these numbers.

A baseline UNKNOWN result is censored. Its elapsed time is time to the cap, not a completed solve, so it receives no speedup ratio. Local-pattern cases use the sequence A, B, A on fresh plain PDs; their wrapper searches before falling back. That deliberately exposes search overhead on cases already decided cheaply by existing filters.

6.3 Rational-pair unknots

Use $w = (AB)^k$, so $m = 2k$ and the displayed unknot has $n = 4m + 4$ crossings. Table 2 gives the measured results. The new method is the complete two-bridge determinant branch on these supplied sources.

n	Old recognition (ms)	New recognition (μ s)	New total (ms)	Paired speedup	Old status
12	3.068	19.79	0.129	147.9×	UNKNOT
20	5.707	22.28	0.263	253.2×	UNKNOT
28	14.926	23.58	0.226	610.9×	UNKNOT
36	56.906	25.48	0.272	1997.1×	UNKNOT
44	66.522*	30.23	0.327	—	UNKNOWN
52	78.735*	30.04	0.381	—	UNKNOWN

Table 2: Fresh-diagram recognition measurements. *New total* includes PD construction. Asterisks mark time to an object cap, not a completed baseline decision. Ratios are paired medians.

The most substantial comparisons are the completed diagonal unknot cases: all ordinary one-sided invariant filters are inconclusive, and the old path reaches the exact Khovanov scan. At 44 and 52 crossings, the new arithmetic branch completes while the explicit old path reaches the object ceiling. This is an improvement in achieved decisions at a fixed budget, not a quantified speedup over an unknown eventual completion time.

The post-elimination object counts in the completed old scans are

$$10, \quad 58, \quad 338, \quad 1970.$$

Their maximum boundary has only four endpoints. For $k = 1, 2, 3, 4$, these counts equal $p_k + q_k$, the graded lower bound in Corollary 3.4. The corresponding counts before elimination are 27, 157, 915, 5333. The numerical agreement supports the interpretation that the retained information itself is costly, rather than merely reflecting missed elementary cancellations. It is not a proof that every later scan is minimal, nor a lower bound for every scan order.

For perspective, at $k = 12$, the mathematical family has a displayed 100-crossing unknot and

$$p_{12} + q_{12} = 2,623,476,242.$$

A full graded complex at that prefix has at least this many explicit summands. This number is a theorem-derived lower bound, verified arithmetically by the diagnostic, not a claim that a billion-object benchmark was run.

6.4 Nontrivial sources and compressed coefficients

The off-diagonal example uses $u = A^8$ and $w = B^8$. Its 36-crossing knot is already rejected by the baseline Rasmussen interval in a median 0.102 ms. The arithmetic stage takes 17.64 μ s,

with paired speedup $5.35\times$; its construction-plus-recognition time is 0.243 ms. This is a useful guard against interpreting every rational example as an exponential fallback case.

Table 3 uses the Alexander-one family (4.4). These knots pass the determinant and Alexander-one tests by construction; the baseline rejects them with its Jones filter. Their arithmetic nontriviality follows from three exceptional fibers.

a	Crossings	Old recognition (ms)	New recognition (μ s)	New total (ms)	Paired speedup
3	15	0.964	20.44	0.148	$46.0\times$
7	47	3.318	22.11	0.311	$166.1\times$
31	575	35.322	40.14	4.973	$977.9\times$

Table 3: Alexander-one, signature-zero, nontrivial Montesinos knots. The large recognition-only ratios exclude source expansion from both recognition timers; the end-to-end column makes that cost visible.

A separate compressed experiment chooses $a = 2^b + 1$, using

$$b \in \{64, 256, 1024, 4096, 16384, 65536\},$$

and calls the arithmetic API without constructing any PD. The resulting certificates all have determinant one and reject via the exceptional-fiber argument. For the last sample, a has 65,537 bits, the expanded crossing count has 131,072 bits, and the median production-certificate time is 1.383 ms. Its maximum tracked arithmetic entry length is 262,144 bits. The certificate verifier is checked outside that production timer. These numbers illustrate the difference between compressed and expanded input models; they do not represent the processing of an explicitly listed diagram with that many crossings.

6.5 Local search: useful certificates, mixed performance

Table 4 records the wrapper that searches first. The first three rows are negative controls for the default six-template catalogue. The last two replace a crossing disk in the Conway PD by an actual forbidden tangle, producing 16- and 25-crossing host diagrams. Those two rows use the supplied relevant template, so they should not be compared to an exhaustive template-discovery procedure.

Case	n	Old (ms)	Wrapper (ms)	Paired speedup
Conway catalogue control	11	0.706	1.197	$0.59\times$
Hard-unknot catalogue control	8	0.444	0.691	$0.64\times$
Trefoil catalogue control	3	0.015	0.065	$0.26\times$
Planted $1/3 + 1/3$	16	0.320	0.276	$1.21\times$
Planted $1/(-3) + 1/5 + 1/7$	25	0.049	0.328	$0.15\times$

Table 4: Local witness discovery is sound but not a universal speed improvement. A ratio below one means the wrapper is slower. The cheap Rasmussen interval already rejects the last host, explaining its regression.

The local certificate is mathematically stronger than merely checking one chosen closure, but that does not make its current placement optimal. Searching before a cheap Seifert or Rasmussen certificate can lose time. The appropriate current release policy is optional search, with future experiments on placement after cheap filters and before expensive scans. The modest planted two-summand improvement should not be overinterpreted: submillisecond timings are noisy, the sample is small, and the control ratios already expose nontrivial variation.

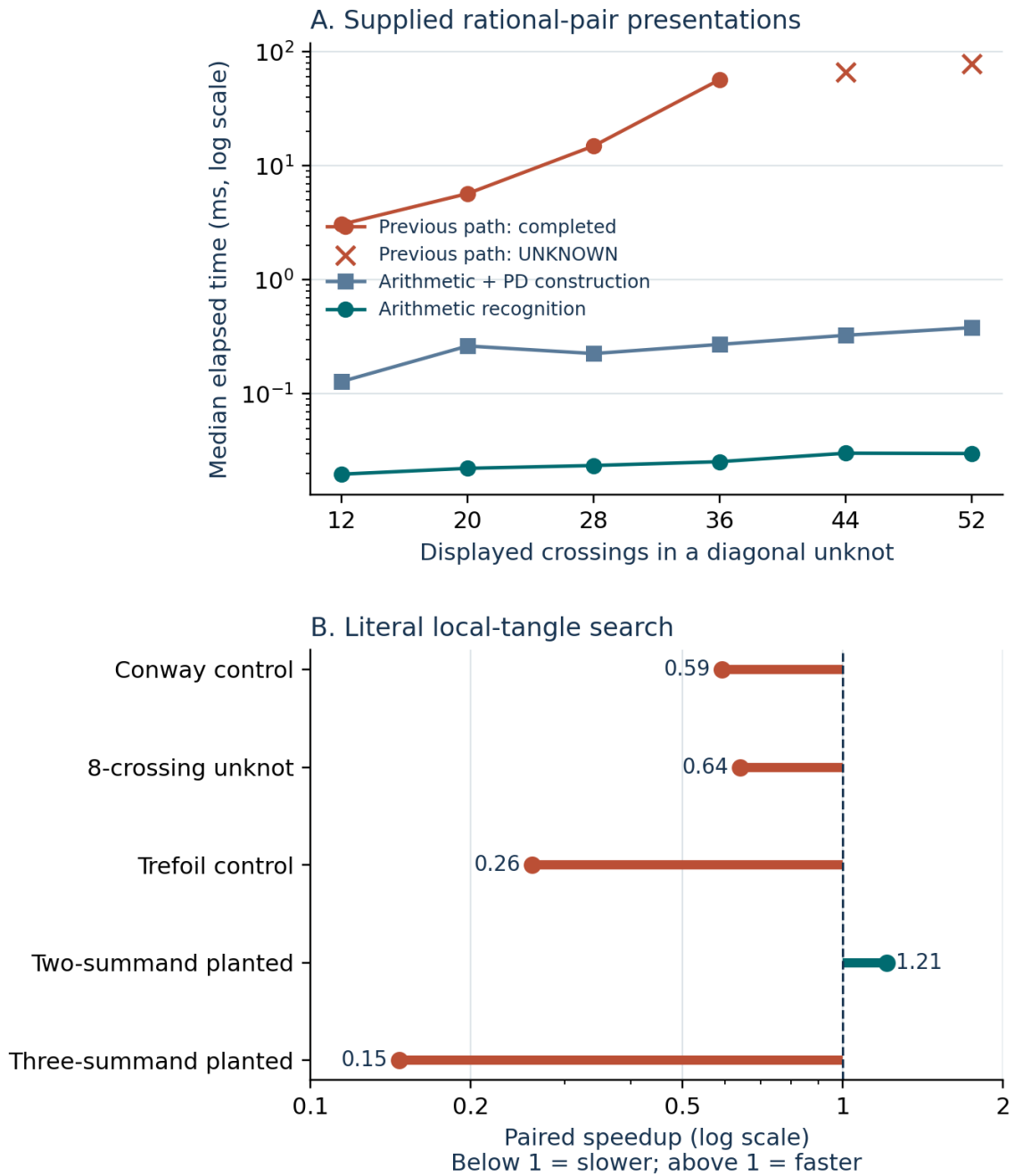


Figure 1: Recorded pipeline behavior. Top: diagonal unknot times, with baseline cap exits marked separately from completed decisions. Bottom: the optional local wrapper’s paired speedups, including all measured regressions. The figure is generated from the supplied JSON, not fitted to an asymptotic model.

6.6 Reproducibility limits and useful follow-up measurements

The archive’s benchmark driver records every timing sample, both baseline result objects and one new-path result per structured round. Local rounds retain all three result objects. Recorded evidence includes methods, available scan statistics, and the baseline before/after control ratio; end-to-end and compressed calls retain their timing samples and stated summaries. It can regenerate the source examples and planted PDs. The plotting script reads the recorded JSON and requires no new timing run. A separate reproducibility driver exposes the quick tests, full suite, mathematical diagnostics, and optional repeated benchmarks.

The measurements establish performance for these sources and this environment. They do not establish average-case behavior on random knots, the effectiveness of discovering hidden Montesinos structure, or the maximum performance of every pre-existing backend. In particular, the primary ablation compares the new arithmetic branch to the default previous path, not to a competition among all backend settings. The useful next comparisons are plain-PD rational extraction, pipeline placement of local search, stronger adversarial source scrambling, independent environments, and accounting for expanded diagram construction. The mathematical quadratic arithmetic bound rests on the proof above, not on fitting the six compressed timing points.

7 A precise route toward quasi-polynomial complexity

The new arithmetic stage is polynomial on its supplied input class. The bounded-description search is quasi-polynomial on a restricted one-sided discovery problem. Neither result bounds the cost of the remaining general instances. The objective now is to identify an interface and a discovery theorem that can survive those instances without materializing the exponential objects isolated above.

7.1 Which compression target should be changed

There are at least four different mathematical contracts for a tangle summary:

1. *Relative-boundary isotopy information.* A rational slope determines a rational tangle up to isotopy fixing its marked boundary. This is exact topological information, compactly encoded.
2. *Full homotopy information.* A Bar–Natan complex retains enough data to compute the homology of every planar composition. Explicit representatives can be exponentially large at the prescribed checkpoints of Section 3.
3. *All Boolean continuation answers.* The relation $P \sim P'$ means that every allowed continuation produces the same unknot decision. Section 2 shows exponentially many classes even for a fixed four-point boundary. It also rules out a small linear factorization of that final Boolean predicate on the constructed family.
4. *A certificate for the actual input.* A Montesinos arithmetic proof or a forbidden local disk can decide one input without constructing all continuation answers. This is the contract used by the delivered early exits.

These contracts should be reflected in types and function boundaries. A local nontriviality witness cannot be inserted as a substitute for a full chain complex at an arbitrary internal node. A rational slope can represent the topology of a rational piece, but this alone does not provide

a fast operation for gluing it to an arbitrary nonrational complex. The required operation must either have its own theorem or retain the original diagram for an exact fallback.

The rank-two identity

$$D_m = ps^\top - qr^\top, \quad H_m = \mathbf{1}\{|D_m| = 1\}$$

illustrates a productive distinction. A low-dimensional arithmetic value can encode an exact answer after a nonlinear readout even when the Boolean connection matrix has full rank. State-count and Boolean-rank obstructions should therefore be used to reject an unsuitable representation contract, not to reject all arithmetic approaches.

7.2 What the proved restricted discovery theorem actually buys

Proposition 4.9 removes the need for a supplied occurrence map on a specified family. It is a genuine algorithm: enumerate bounded sources, reject arithmetic or component failures, compile a literal plane pattern, and propagate each possible root image. When $h = O(\log n)$, its running time is $n^{O(\log n)}$, including bit arithmetic and the polynomial work per candidate.

The quantifiers are the important part. Completeness holds for forbidden tangles that already appear as literal admissible diagrams with the specified source bound, orientation, and four distinct external edges. The algorithm does not search arbitrary isotopies. It does not prove that every nontrivial diagram contains a short witness of this kind. It also never certifies an unknot merely because no forbidden template was found.

The hypothesis is substantial. A continued fraction with a long sequence of coefficients equal to two can have $h = \Theta(n)$; storing its slope still takes only $O(n)$ bits, but the enumeration theorem's logarithmic coefficient-count hypothesis need not hold. Conversely, a large twist coefficient can contribute many displayed crossings while occupying one coefficient slot. This explains why coefficient count is useful and why it is not a universal measure of diagram difficulty.

The fixed catalogue already gives a concrete scalable rejection family with arbitrary exteriors: every accepted literal occurrence of $R(1/3)+R(1/4)$, for example, proves nontriviality regardless of the number or complexity of exterior crossings. A useful next theorem would enlarge this family through certified local rewrites or a rational-region extractor while preserving a proved total discovery bound.

7.3 A hierarchy accounting criterion

The geometric hierarchy route is complementary to the arithmetic tangle route. It should be audited using the iteration count and operation cost as separate parameters.

Proposition 7.1 (Conditional accounting, not a new hierarchy theorem). *Suppose a fully specified implementation has at most $L(g+1)^L$ elementary hierarchy events, as in the iteration estimate of [4, Proposition 9.1]. Suppose additionally that*

$$g \leq n^a, \quad L \leq b(\log(n+2))^c$$

for fixed constants $a, b, c > 0$, and that all encoding growth and bit operations at each event are bounded by $\exp(O((\log(n+2))^{c+1}))$. Then its total bit time has the same form

$$\exp(O((\log(n+2))^{c+1})).$$

In particular, $c = 1$ yields $n^{O(\log n)}$, whereas $c = 2$ yields $n^{O((\log n)^2)}$.

Proof. Taking logarithms of the event bound gives

$$\log L + L \log(g + 1) = O((\log(n + 2))^{c+1}).$$

Multiplying the number of events by the assumed per-event bound adds two quantities of that order in the exponent. This accounts for both iteration count and operation cost. It proves no one of the hypotheses. $\square$

This accounting is useful precisely because each hypothesis can fail independently. A short hierarchy does not ensure short coordinate descriptions. A combinatorially decreasing potential does not bound the cost of finding the next compression. Replacing a surface by a compressed description does not make intersection and update operations efficient without an explicit algorithm. These obligations are central to turning the hierarchy programme into a reviewed implementation.

The present tangle lower bounds do not refute such a geometric approach. They concern specific algebraic summaries. Their practical lesson is that geometric events should not be forced through an explicit full complex merely to preserve an interface whose information is richer than the decision requires.

8 Further research questions and concrete milestones

The following questions are ordered roughly from immediately actionable implementation work to broader mathematical objectives. Each specifies what would count as progress. They are research proposals, not assumptions used to justify the existing code.

Research question 1 (Certified extraction of visible rational regions). Can a validated PD be reduced to maximal visible rational regions in $O(n \log n)$ or another explicit polynomial time, while returning relative-boundary isotopy certificates and exact slopes?

Start with bigon chains and the inverse operations of the rational compiler, retaining cyclic port order, crossing signs, and a replayable contraction tree. A useful first result would recognize the supplied diagonal family after source metadata is removed and after arbitrary relabeling and crossing reordering. A stronger result would admit carefully specified flypes or local isotopies. The theorem should define which diagrams the extractor is complete on, and charge each graph update rather than assuming repeated scans are free.

Research question 2 (Choosing the placement of local search). Can local obstruction search run after all cheaper certificates and before the expensive fallback with a small, provable overhead bound?

The current negative controls already demonstrate why this matters. Measure construction, validation, compilation, discovery, and fallback separately, using held-out hosts as well as planted patterns. Share target dart arrays and precompiled templates. A useful release gate is an end-to-end improvement on difficult plain PDs while bounding overhead on trefoils, descending diagrams, and homogeneous examples. A benchmark selected only from successful local witnesses would not answer the question.

Research question 3 (Practical bounded-description enumeration). Can the $n^{O(h)}$ enumeration be pruned substantially without losing completeness for its stated literal class?

Possible exact pruning keys include expansion size, endpoint pairing, crossing-sign counts, rooted ribbon neighborhoods, and arithmetic residues. Canonicalization must respect the

difference between two isotopic rational sources and two different literal diagrams: equality of slopes is enough for isotopy, but not necessarily enough for an unmodified dart match. An implementation should return the same existence answer as the exhaustive small- h reference enumerator before any asymptotic speed claim is made.

Research question 4 (A larger certified local-tangle language). Which arborescent or graph-manifold tangle classes admit compact peripheral data and complete tests for whether an unknotting complement can exist?

The Montesinos criterion uses more than homology order: its congruence condition excludes $R(1/3) + R(1/4)$ even though the numerator and denominator determinants are coprime. An extension should retain the exact boundary slopes, exceptional-fiber information, and marking needed for gluing. It should classify rational and product degeneracies explicitly. Merely recording the first homology group of each piece loses the peripheral information that makes the current obstruction effective.

Research question 5 (Typed arithmetic composition with an arbitrary exterior). What exact interface permits a rational slope or a compact Seifert piece to interact with a nonrational remainder without expanding its full complex?

A first milestone could support one rational piece glued to a small explicit remainder with a polynomial cost in its bit description and the remainder's actual size. The correctness theorem must specify which topological or homological answer the interface preserves. There should be explicit conversion and fallback rules. A compact slope object is not automatically an object of the pre-existing chain-complex category.

Research question 6 (Continuation-dependent processing of the Pell family). Can a general PD strategy detect and exploit the relationship between prefix and actual suffix before it reaches the exponential checkpoint?

On the diagonal family, the decisive relation is the unimodular pairing $p_k s_k - q_k r_k = -1$. An algorithm that deliberately constructs the full prefix complex pays for information that the suffix immediately collapses. A useful experiment would compare symmetric gluing, look-ahead, and arithmetic extraction on source-erased diagrams with controlled perturbations. A useful theorem would bound the size of all intermediate representations for that algorithm, rather than only bounding the final unknot rank.

Research question 7 (Compact homological data beyond multiplicities). Can differentials and extension data for structured tangle complexes be stored in an arithmetic circuit, recurrence, or other succinct form on which the needed closure operations remain efficient?

The lower bound permits such representations because it counts expanded summands. It does not prove they exist with efficient operations. A scalar multiplicity alone omits differential data; capping a multiplicity can be unsound if later maps interact with its copies. The target should include composition, cancellation, restriction to a justified homological window, and exact rank decisions on the permitted output, with explicit bit-growth bounds. Symbolic expression size must be analyzed alongside the number of graph nodes in the representation.

Research question 8 (Useful arithmetic values before a nonlinear readout). Which intermediate arithmetic invariants have a low-dimensional gluing law but a sufficiently strong exact terminal predicate?

The signed determinant pairing is a complete example on rational continuations. It converts a rank-two arithmetic matrix into a full-rank Boolean matrix by testing membership in $\{1, -1\}$.

For more general pieces, one might study peripheral matrices, presentations with controlled relations, or exact equations for distinguished fillings. An arithmetic invariant that is not complete should remain a one-sided filter. In particular, a compact Jones or Euler calculation should not be promoted to a general unknot detector without an additional theorem.

Research question 9 (Generalized representation lower bounds). Can the continuation construction distinguish other proposed summary models, including bounded-degree nonlinear readouts, compressed modules, or restricted arithmetic circuits?

Any proposed lower bound must specify the model and its input encoding. The current identity matrix already implies that a representative-subfamily scheme preserving the existence of an accepting prefix for every displayed suffix must retain all 2^m prefixes: suffix Q_w accepts only P_w . This does not prevent a symbolic set representation. A stronger useful result would identify exactly which operations force expansion and which ones preserve compactness, rather than claiming a general complexity separation from a state count.

Research question 10 (Sharper, simpler independent verification). Can the separate Montesinos verifier be reduced from the conservative cubic bound to $O(B^2)$, or better with fast arithmetic, without merely reusing the production implementation?

The nested replay exists to check tracked prefix statistics, not to decide the topology. One option is to remove nonessential statistics from the mathematical certificate and verify them separately. Another is to maintain independent prefix products or use a product tree. The important outcome is an independent small checker with a transparent source-to-fraction and fraction-to-verdict proof, not a second function that trusts the producer.

Research question 11 (Short positive certificates and formal proof kernels). Can a determinant-one arithmetic decision be turned into a compact, replayable topological certificate and formalized in the ProveIt setting?

For compressed sources, a fully expanded Reidemeister movie may be longer than the source by an exponential factor. A suitable certificate should instead record rational-tangle identities, Euclidean arithmetic, and boundary-preserving gluing steps in compressed form. Its checker must verify all signs and primitive pairs. For local negative certificates, the disk lemma can be formalized as a statement about finite dart permutations and a plane embedding, with the imported topological theorem kept explicit. Formalizing the disjoint-cone and triangle-representation lemmas would give smaller initial proof-assistant milestones.

Research question 12 (Extending the local certificate geometry). Can the implementation safely admit two ports on a single exterior edge, disconnected crossing shadows with simple arcs, or additional boundary marking conventions?

The initial four-distinct-edge contract avoids these cases and is sound. An extension needs a new explicit cut-point representation and a proof that the accepted common face still bounds the intended disk with no hidden components. Disconnected shadows require more than simply reusing root propagation from one component. Separate topology lemmas and adversarial tests should precede any expansion of the accepted class.

Research question 13 (A useful structural completeness theorem). Can every diagram in a broad, explicitly specified class be transformed or decomposed so that arithmetic classification or a short forbidden local certificate becomes available within quasi-polynomial cost?

Candidate classes should be defined by their diagrams and allowed moves, not retrospectively by whether the implementation succeeds. One could begin with a controlled arborescent

grammar or diagrams with a supplied decomposition tree and then study discovery. Both the transformation count and the size of each intermediate diagram or compressed source must be bounded. The current results provide neither a universal $O(\log n)$ coefficient bound nor a theorem that local forbidden tangles cover all nontrivial knots.

Research question 14 (Hierarchy operation costs and the logarithmic exponent). Can the geometric hierarchy programme supply executable bounds for compression discovery, parallelity-bundle updates, boundary-pattern simplification, and terminal checks, with controlled coordinate growth?

The accounting in Proposition 7.1 identifies the necessary parameters. An implementation audit should attach an explicit representation and cost to each operation from the mathematical algorithm. It should separately establish whether the needed hierarchy length is $O(\log n)$ or $O((\log n)^2)$, since those produce different quasi-polynomial exponents under polynomial pattern complexity. A declining hierarchy potential alone is insufficient.

Research question 15 (Adversarial families and fair backend comparisons). Which examples remain hard after provenance is removed, cheap certificates are applied, and the best supported existing backend is selected?

The current diagonal family isolates a real expanded-complex problem; the Alexander-one family isolates the failure of a common invariant; the planted local hosts test certificate discovery. A broader study should separate these purposes and include source-erased rational diagrams, hidden local patterns, examples with no matches, and alternate compositions. Keep resource exits censored, use fresh caches, report process startup when relevant, and compare end-to-end time as well as inner kernels. This would guide algorithm design without confusing favourable source metadata with a general discovery improvement.

8.1 Recommended sequence of follow-up contributions

The next contribution should implement and certify visible rational-region extraction, then benchmark the present arithmetic decision on ordinary source-erased PDs. That would remove the largest practical qualification from the supplied-source speedups on an explicit class.

A second contribution should move or schedule local search using the measured pipeline costs, add an exact small- h reference enumerator, and develop completeness-preserving pruning. Its claim should remain a local-witness discovery bound until a wider structural theorem is proved.

A third contribution should implement one typed arithmetic/exterior composition operation with a full correctness proof and a proved bound on all intermediate representations. The Pell checkpoint family gives a sharp test: success should avoid the expanded $p_k + q_k$ cost, not simply hide it in a later conversion step.

The broad quasi-polynomial objective then requires a coverage or hierarchy theorem together with executable operation costs. The delivered arithmetic and local certificate stages are exact components available for that programme. The lower bounds identify a representation choice that can defeat the programme even on unknots, while the arithmetic construction demonstrates a concrete way around it in a mathematically complete case.

A Algorithm summaries and size accounting

The following pseudocode fixes the arithmetic decisions independently of the language-specific implementation. Fractions remain primitive because each recurrence is unimodular. The

integer accumulator is deliberately not reduced as a rational number.

```

PrimitiveFraction([a0, ..., ak]):
    (p, q) := (1, 0)
    for a in [ak, ..., a0]:
        (p, q) := (a*p + q, p)
    reject unsupported final infinity if q = 0
    if q < 0: (p, q) := (-p, -q)
    return (p, q)

MontesinosDecision(e, source_blocks):
    exceptional := []
    for block in source_blocks:
        (p, q) := PrimitiveFraction(block)
        (a, beta) := EuclideanDivmod(p, q)
        e := e + a
        if q > 1:
            append (beta, q) to exceptional
    (D, Q) := (e, 1)
    for (beta, alpha) in exceptional:
        (D, Q) := (D*alpha + beta*Q, Q*alpha)
    reject input outside the knot domain if D is even
    if len(exceptional) >= 3:
        return KNOTTED, exceptional-fiber certificate
    if abs(D) = 1:
        return UNKNOT, two-bridge certificate
    return KNOTTED, determinant certificate

```

The local obstruction checker uses the unnormalized primitive slopes as well:

```

ForbiddenLocalSource(source):
    require integer term e = 0
    require at least two finite nonintegral rational summands
    compute primitive (pi, qi), with qi > 1
    require two open strands and no closed component
    if number_of_summands >= 3: return FORBIDDEN
    Q := q1*q2
    F := p1*q2 + p2*q1
    return (F mod Q) not in {1, Q - 1}

FindLiteralWitness(host, source):
    compile and verify connected plane pattern with four ports
    require ForbiddenLocalSource(source)
    for target_root in crossings(host):
        for offset in {0, 2}:
            propagate the unique possible map over internal edges
            reject collisions, inconsistent rotations, or overpair errors
            check every remaining dart and four distinct outgoing edges
            if all source and separating-disk checks pass:
                return source, crossing map, rotations
    return UNKNOWN

```

For the expansion row, even a conservative implementation that revisits the current explicit graph at each construction step performs polynomially many operations in $B + C$. The delivered compiler avoids unnecessary whole-graph mirroring, but this report makes no finer worst-case compiler claim than is needed. The production arithmetic bound applies before any

Operation	Bound	Input and scope
Two-twist rational summary and gluing	$O(m^2)$ bits of time	Words of length m ; $O(m)$ bits of arithmetic data
Montesinos production arithmetic	$O(B^2)$ bits of time	Complete compressed source length B , including delimiters
Independent branch verification	$O(B^3)$ bits of time	Valid $O(B)$ -size certificates; parse purported input as well
Literal PD construction	Polynomial in $B + C$	Must charge C expanded crossings; cap checked before allocation
One compiled pattern search	$O(nm)$ word operations	Connected m -crossing pattern, indexed target already validated
Public finite-catalogue wrapper	$O(n \log(n + 2) + nM)$ word operations	Total expanded pattern size M , plus all source/compilation costs
Bounded-description discovery	$(n + 2)^{O(h)}$ bit operations	Literal admissible sources with at most h coefficients

Table 5: Complexity statements and the input sizes they actually charge. The abstract matching word bounds have indexed-array realizations. The Python implementation’s table operations use the usual table model.

expansion and is the bound relevant to exponentially large twist coefficients.

B Artifact map and reproducibility

The main article driver is `article/unknot_arithmetic_continuations.tex`. It is assembled from numbered editable sections by `article/build_article.py`; all prose, theorems, tables, and bibliography are included directly in the combined driver. The optional plotted figure is used when its file exists, so the standalone combined source can also compile without the figure assets. The archive contains both the assets and a portable plotting script.

Path in the archive	Purpose
<code>article/</code>	Combined and section TeX, PDF, build script, supplementary BibTeX, and figure assets
<code>fast/</code>	Runnable updated recognizer, standard-library implementation, tests, examples, certificates, and preserved license
<code>integration/</code>	Reviewable patch, application instructions, source manifest, and patch byte-verification manifest
<code>experiments/</code>	Corpus generator, paired benchmark, independent arithmetic and topology diagnostics, reproduction launcher
<code>examples/</code>	Complete structured and planted plain-PD inputs
<code>certificates/</code>	Complete local source-plus-map certificates
<code>results/</code>	Raw timings, finite verification outputs, full-suite and patch-application logs
<code>RESEARCH_PROVENANCE.md</code>	Baseline, imported results, independently derived claims, and limited inspiration attribution

Table 6: Files supplied for review and later integration into ProveIt.

The reproduction launcher accepts an explicit output directory and creates an isolated

replay workspace, leaving the bundled measurement record unchanged. From any working directory, invoke its absolute path with `--quick` for the added suites and finite diagnostics, `--full-tests` for the full suite, `--identity` for all identity matrices through $m = 10$, or `--benchmarks` for a fresh seven-round timing run. The `--all` mode intentionally performs the expensive reproductions as well. Exact examples are in the top-level README.

The repository patch was created for the pinned baseline, not for an unseen future revision. Its manifest permits a reviewer to compare the expected bytes before application. Subsequent repository changes may require a normal patch merge and another targeted validation; that is an integration issue rather than a change in any theorem above.

References

- [1] V. Reshetnikov and contributors. *ProveIt: Topology/UnknotRecognition*. Repository source and research reports, commit 86f23490006c880cd01db9bec10761d39c6395c8, inspected 8 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/86f23490006c880cd01db9bec10761d39c6395c8/Topology/UnknotRecognition>.
- [2] P. B. Kronheimer and T. S. Mrowka. *Khovanov homology is an unknot-detector*. arXiv:1005.4346, 2010. <https://arxiv.org/abs/1005.4346>.
- [3] M. Lackenby. *Unknot recognition in quasi-polynomial time*. Oxford lecture slides, March 2021, especially slides 7 and 25–34. The displayed announced bound is $2^{O((\log n)^3)}$. <https://people.maths.ox.ac.uk/lackenby/quasipolynomial-talk-oxford-compressed.pdf>.
- [4] M. Lackenby. *Incompressible surfaces, hierarchies and unknot recognition*. arXiv:2607.23350v1, 25 July 2026, especially Sections 9–10. <https://arxiv.org/abs/2607.23350>.
- [5] L. H. Kauffman and S. Lambropoulou. On the classification of rational tangles. *Advances in Applied Mathematics* 33 (2004), 199–237. arXiv:math/0311499. <https://arxiv.org/abs/math/0311499>.
- [6] L. H. Kauffman and S. Lambropoulou. On the classification of rational knots. *L’Enseignement Mathématique* 49 (2003), 357–410. arXiv:math/0212011. <https://arxiv.org/abs/math/0212011>.
- [7] L. H. Kauffman and S. Lambropoulou. *Hard unknots and collapsing tangles*. arXiv:math/0601525v5, 26 July 2014; original preprint 2006. Especially Theorems 4–5 and the rational fraction conventions. <https://arxiv.org/abs/math/0601525>.
- [8] L. H. Kauffman and S. Lambropoulou. *From tangle fractions to DNA*. arXiv:math/0311511, 2003, especially Theorem 1 and Section 4.1. <https://arxiv.org/abs/math/0311511>.
- [9] J. M. Nogueira and A. Salgueiro. On embeddings of 2-string tangles into the unknot, the unlink and split links. *Topology and its Applications* 305 (2022), 107866. arXiv:2110.15645v1, especially Proposition 4.4 and Theorem 4.9. <https://arxiv.org/abs/2110.15645>.
- [10] Y. Belousov. *Explicit formulas for the Alexander polynomial of pretzel knots*. arXiv:2502.10370v4, 9 March 2026. *Bulletin of the Australian Mathematical Society*, doi:10.1017/S0004972726101439. Especially Theorem 1(3) and Section 2.3. <https://arxiv.org/abs/2502.10370>.
- [11] D. Bar–Natan. Khovanov’s homology for tangles and cobordisms. *Geometry & Topology* 9 (2005), 1443–1499. arXiv:math/0410495, especially Theorem 2 and Section 7. <https://arxiv.org/abs/math/0410495>.
- [12] M. Khovanov. Patterns in knot cohomology I. *Experimental Mathematics* 12 (2003), no. 3, 365–374. arXiv:math/0201306, especially Section 3. <https://arxiv.org/abs/math/0201306>.

- [13] A. N. Shumakovitch. Torsion of the Khovanov homology. *Fundamenta Mathematicae* 225 (2014), 343–364. arXiv:math/0405474v2, 19 June 2018, Corollary 3.2.C.
<https://arxiv.org/abs/math/0405474>.
- [14] OpenAI. *An exponential two-way deterministic state lower bound for one-way liveness*, inspected source section on relation products and context distinguishability. In the `openai/math` repository, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, 2026. Used only as inspiration for the standard context-separation argument; no headline automata theorem is assumed here.
<https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a>.
The precise source path and blob hash are in the accompanying provenance.